# Classification Trees

# Classification Trees

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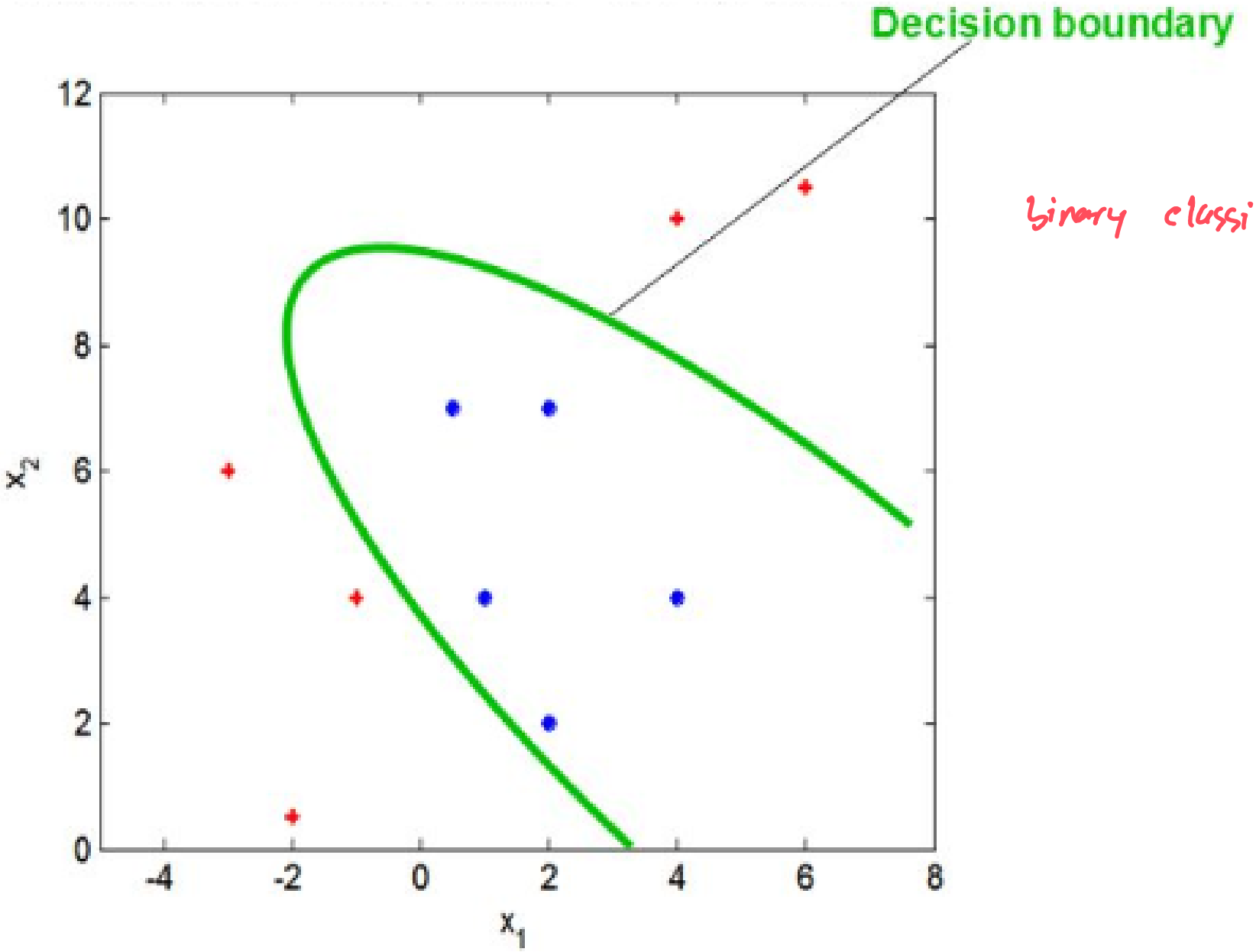
# Reading Materials

- Max Kuhn. Chapter 14. Section 14.1

# Decision Boundary in Classification

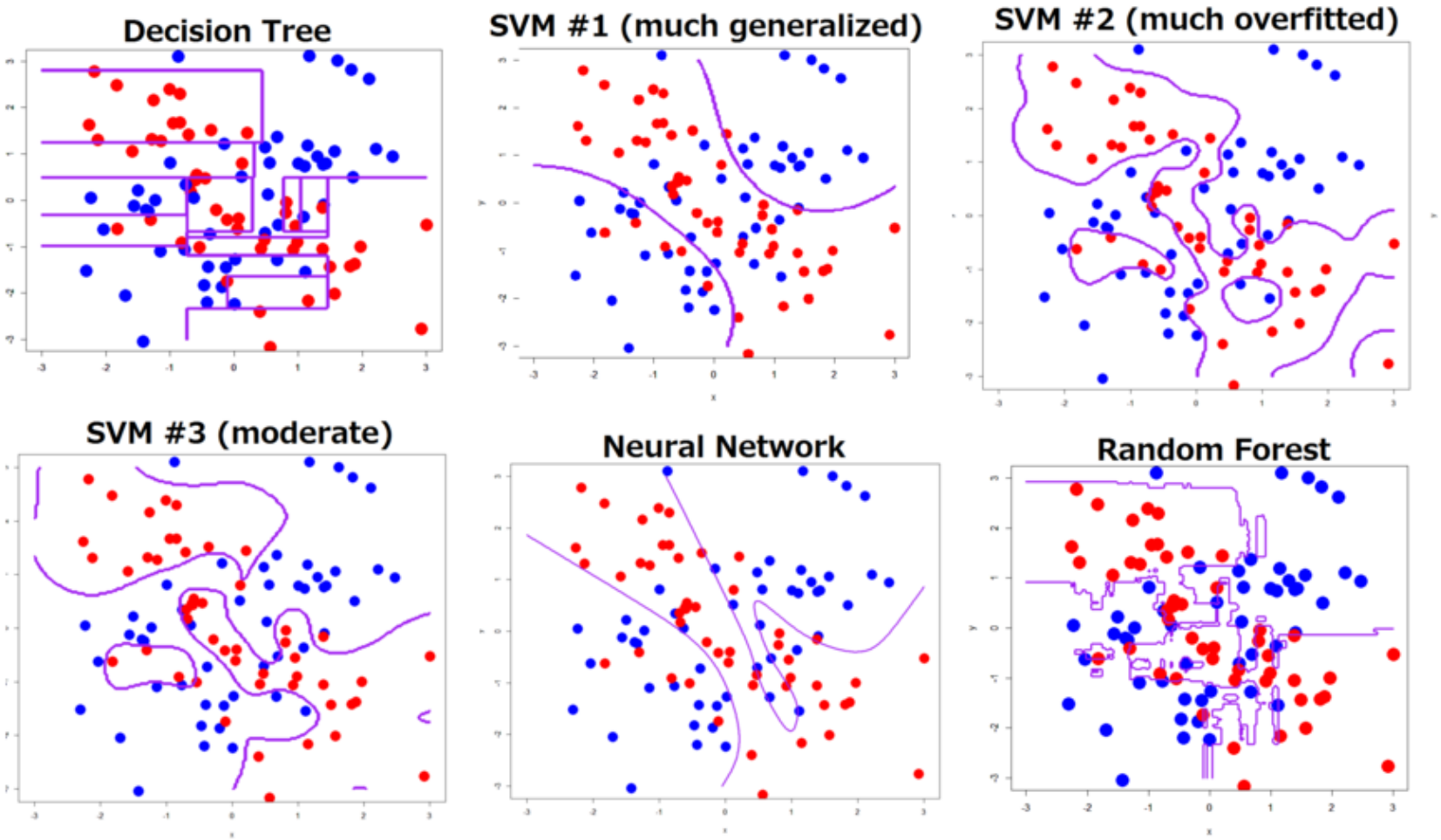
Classification is a process of finding the **decision boundary** that best separates two classes

# Decision Boundary in Classification




binary classification

# Decision Boundary in Classification




- SVM = Support Vector Machine

# Decision Tree

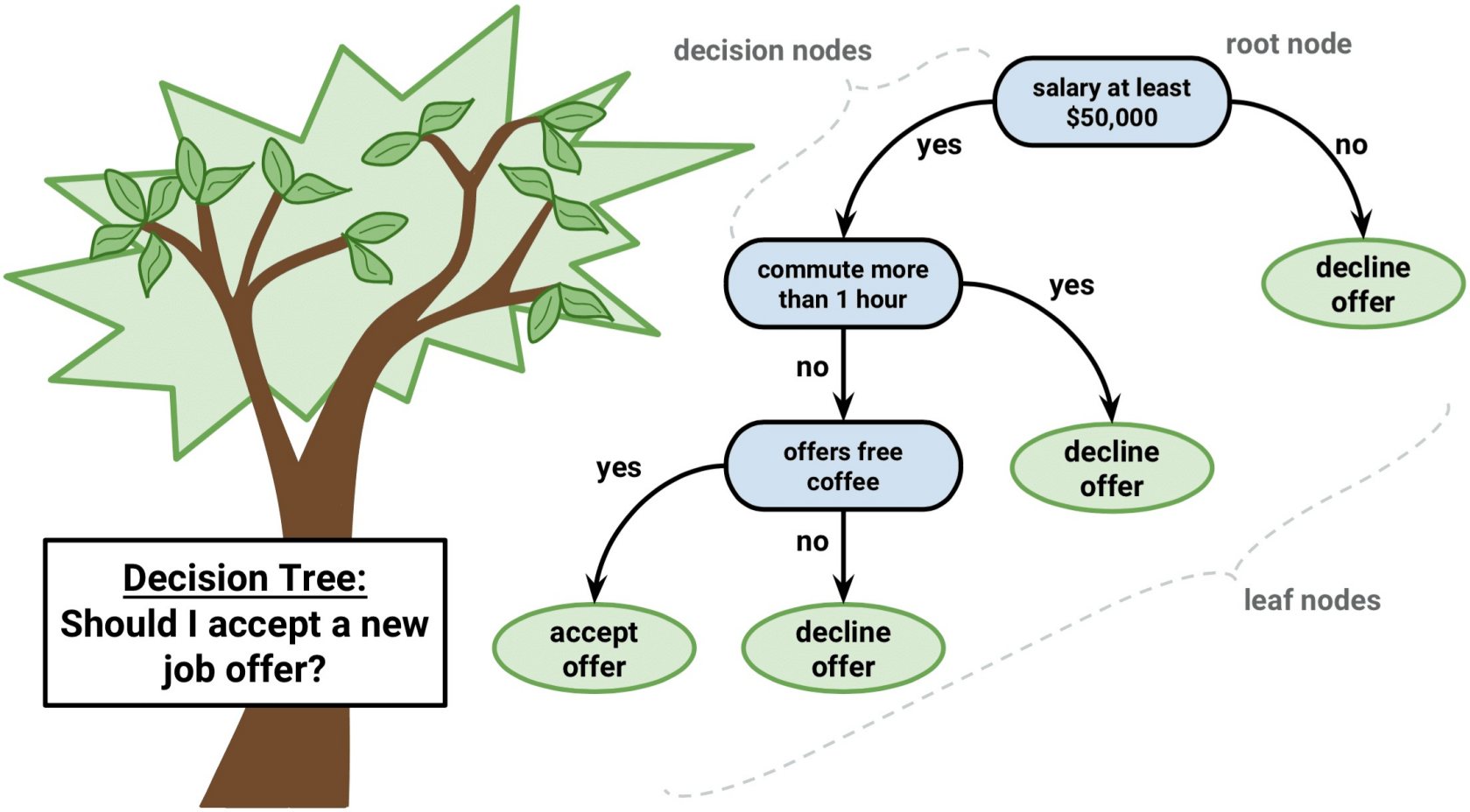
- Decision Tree for classification is **Classification Tree**
- Decision Tree for Regression is **Regression Tree**

# Example of Classification Tree

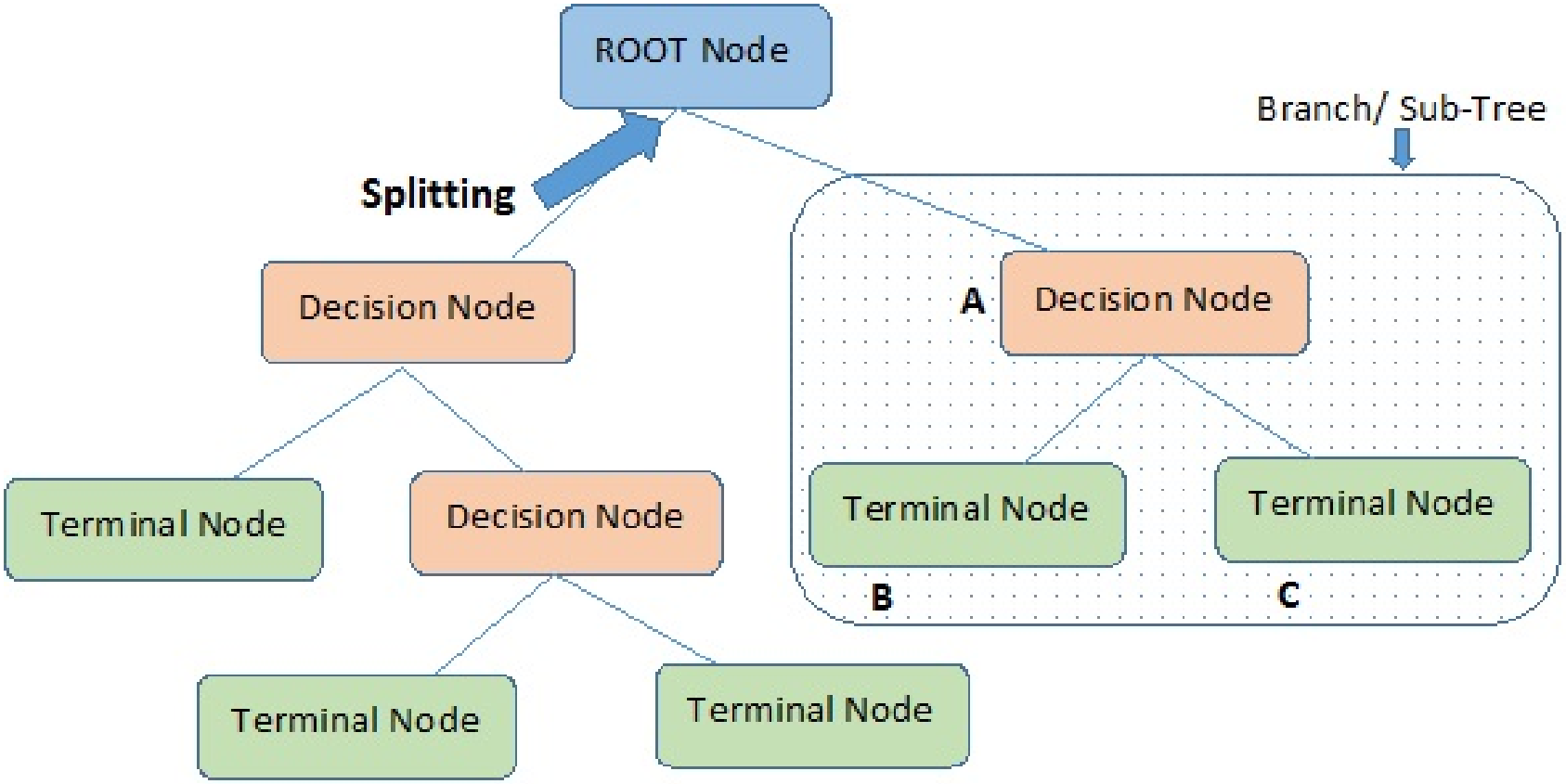
Link

# Example of Classification Tree

binary tree



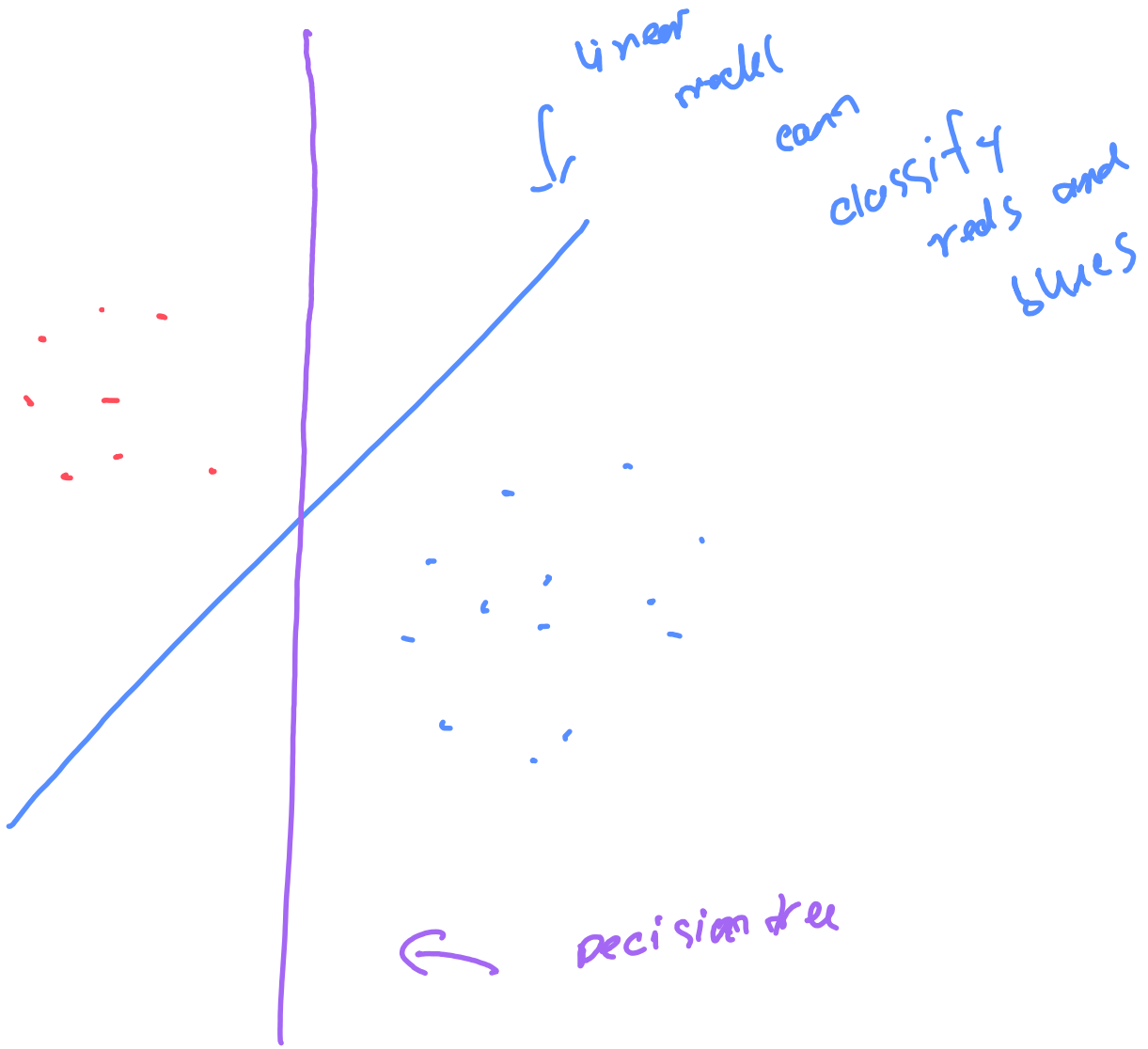
# Example of Classification Tree




**Note:-** A is parent node of B and C.

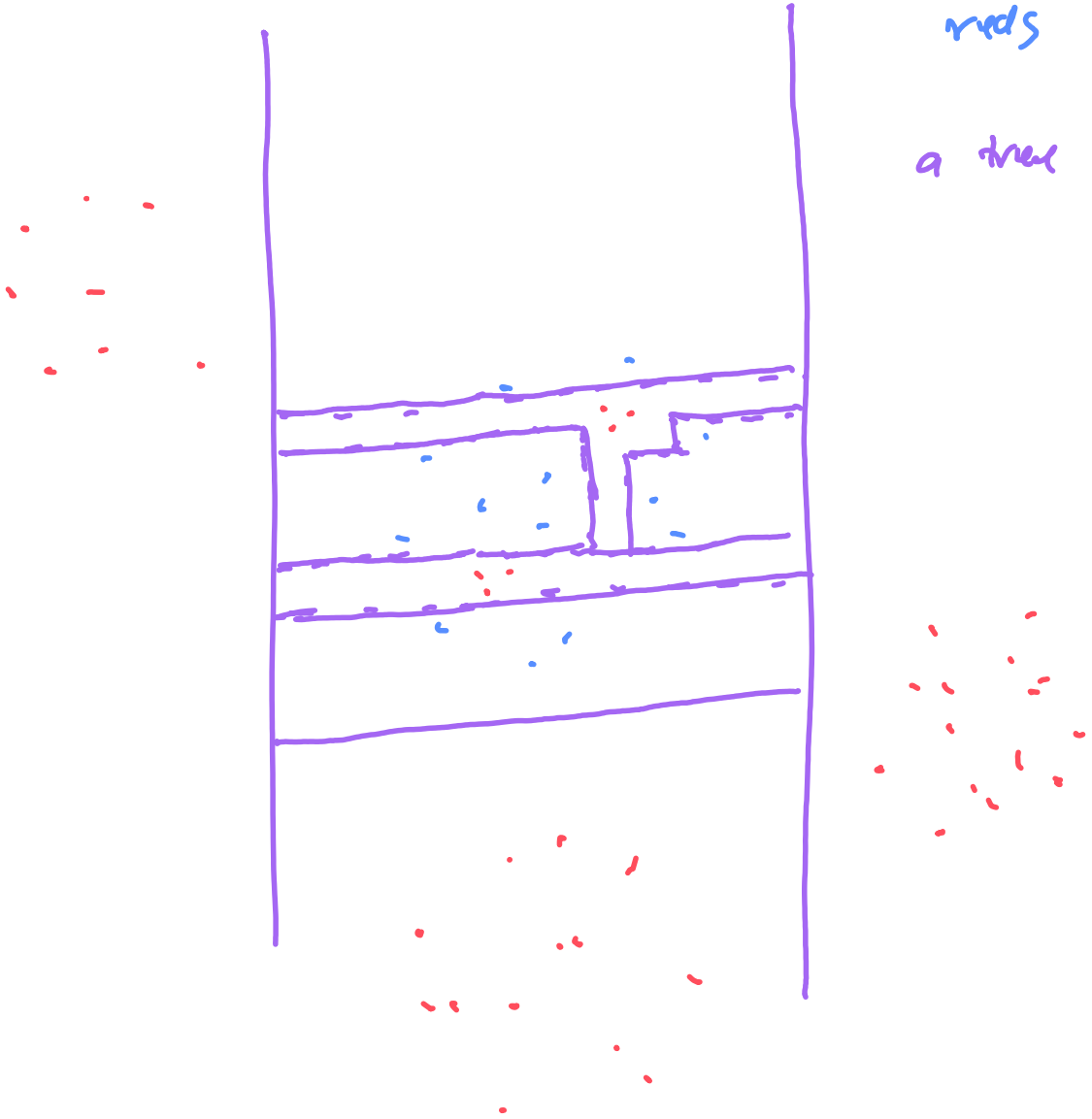
# Classification Tree

- In two dimension, classification Tree's decision boundary is a collection of horiontal and vertical line

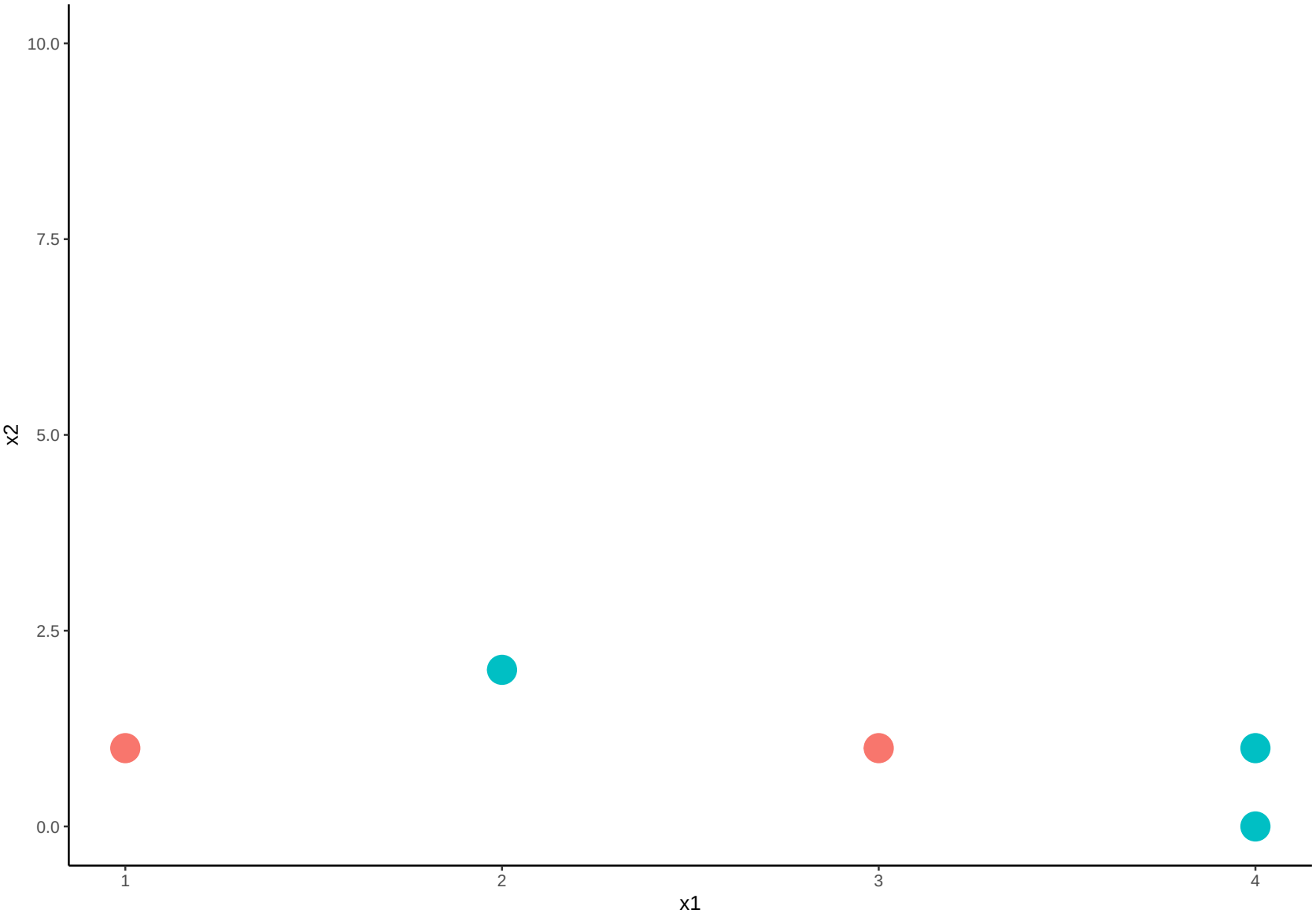

linear model can classify reds and blues
decision tree

Linear model cannot classify reds and blues

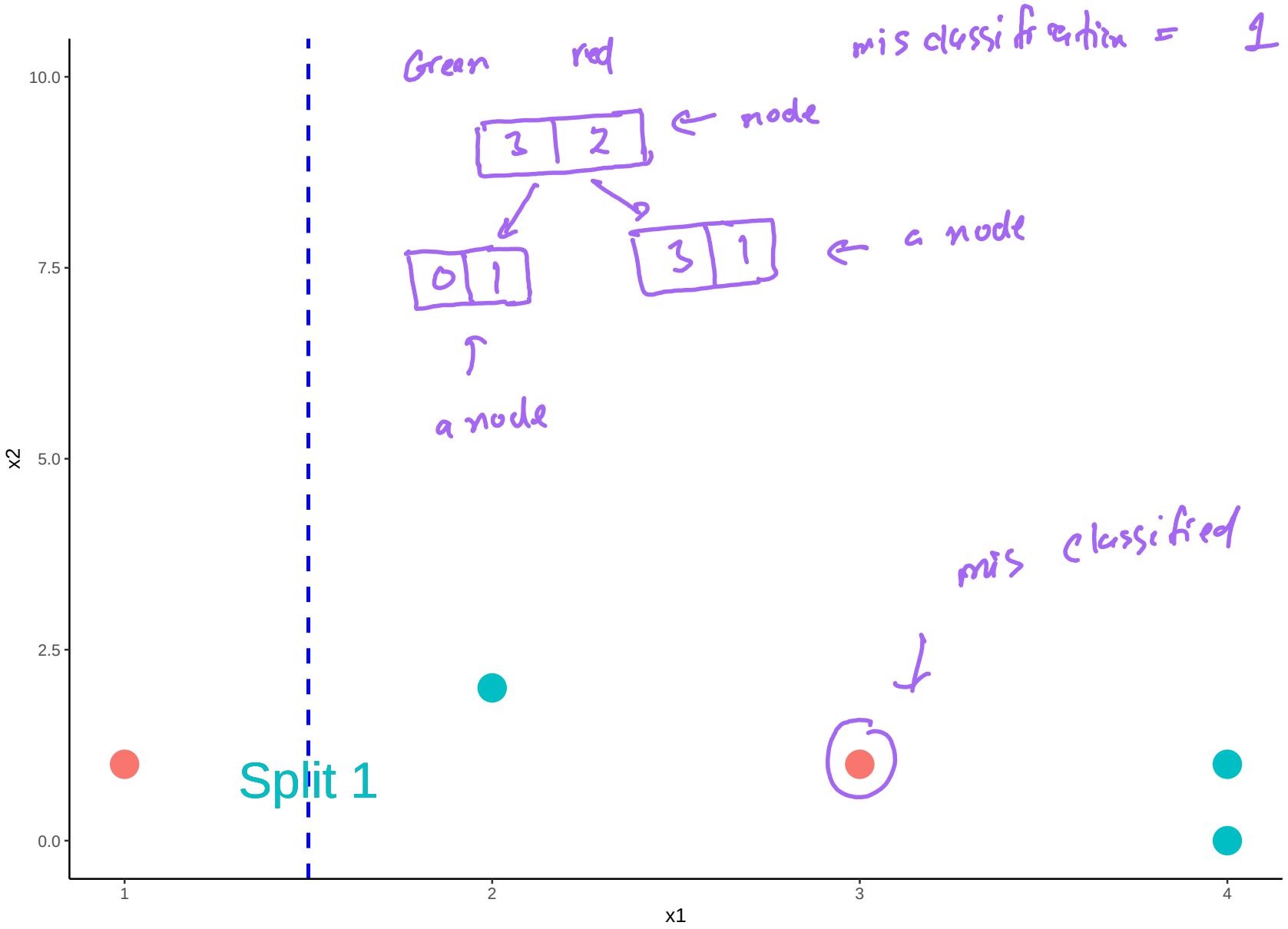
a tree can do that



# Find a vertical line that best seperate **red** and **green**.

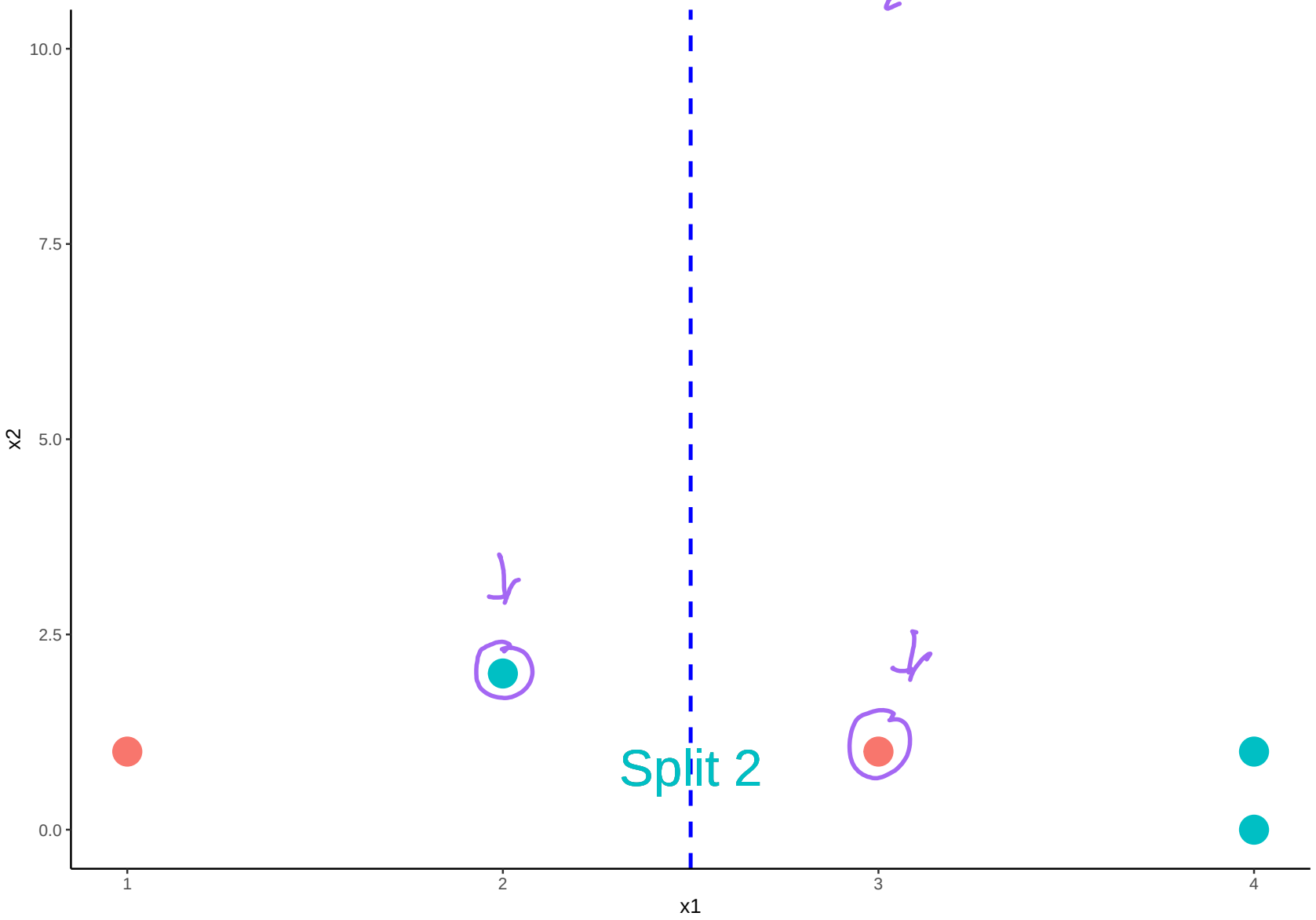


# One way to seperate the reds and greens

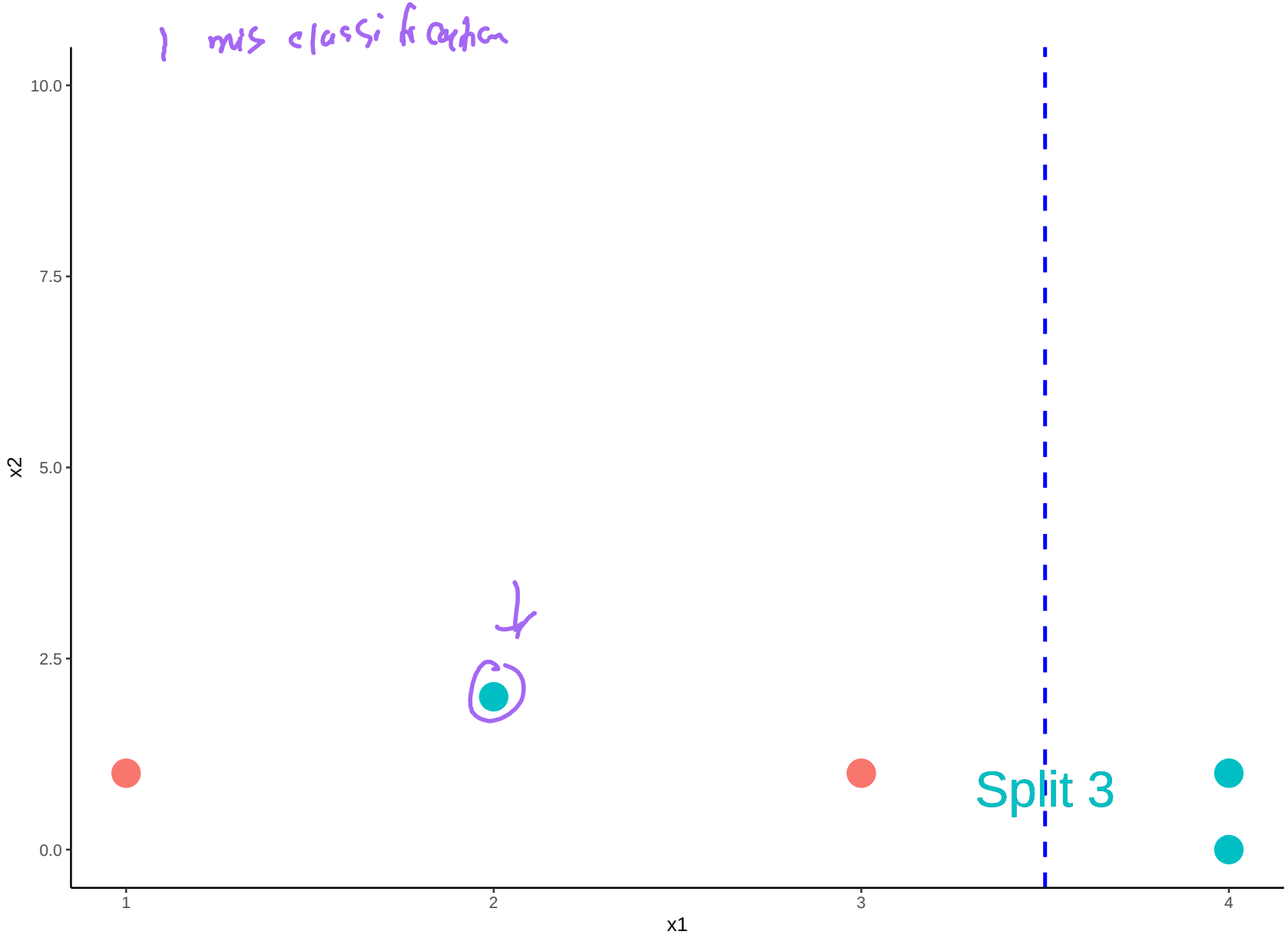


# One way to seperate the reds and greens

2 mis classification



# One way to seperate the reds and greens



# Question

- **Question**: Which is the best split?

# Partial Answer

- It looks like Split 1 and 3 are better than Split 2 since it misclassifies less

# Partial Answer

- Which is the better split between Split 1 and Split 3?

# Partial Answer

- We need to find a way to measure *how good a split is*

# Impurity Measure

- The impurity of a node (**a node = a subset of the data or the original data**) measure how uncertain the node is.

Impurity $\approx$ uncertain (unsure)

Impurity $\left( \boxed{1 \mid 99} \right) < \text{imp} \left( \boxed{40 \mid 60} \right)$

# Impurity Measure

- For example, node A with 50% reds and 50% greens would be more uncertained than node B with 90% reds and 10% greens. Thus, node A has greater impurity than node B.

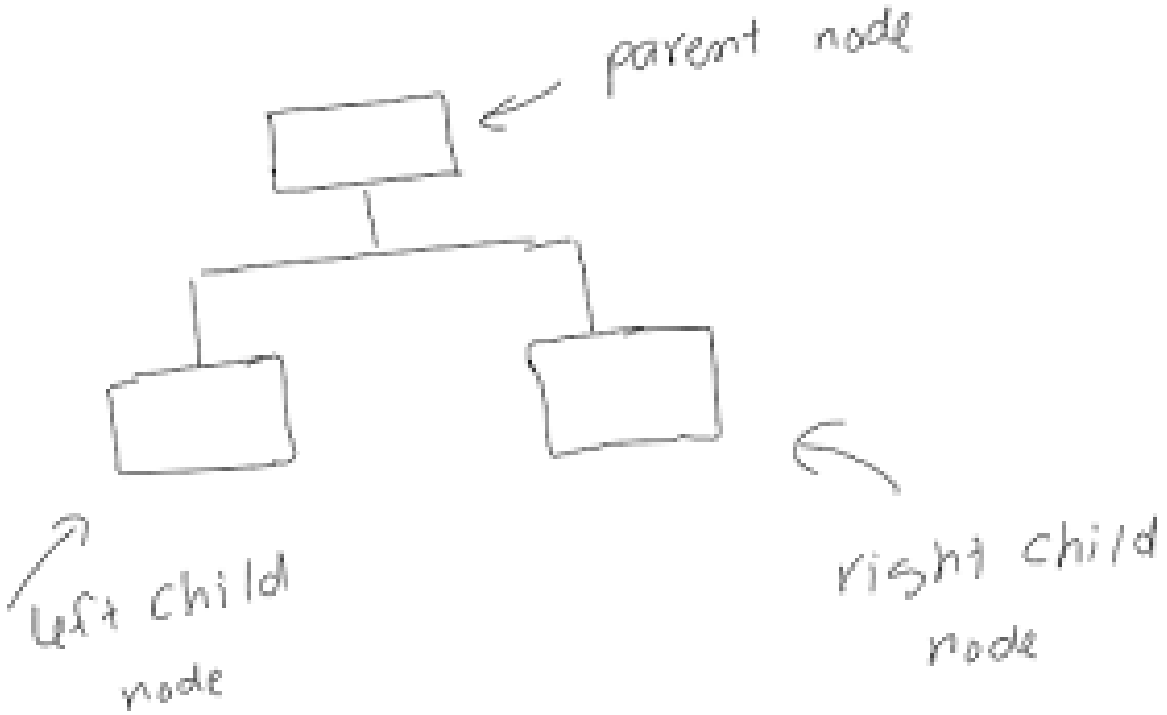
# Impurity Measure

- More uncertained = Greater impurity

# Children Impurity

- A split resulting smaller children impurity is a **better split**

# Children Impurity ($I_{children}$)




$$I_{children} = \frac{N_{left}}{N} I_{left} + \frac{N_{right}}{N} I_{right}$$

- $N_{left}$ and $N_{right}$ are the number of points in the left child node and right child node, respectively.
- $N_{left} + N_{right} = N$

# Impurity Measure

- Impurity can be measured by: classification error, Gini Index, and Entropy.

# Impurity Measure

- Let $p_0$ and $p_1$ be the proportion of class 0 and class 1 in a node.

$$\text{By Classification Error: } I = min\{p_0, p_1\}$$

$$\text{By Gini Index: } I = 1 - p_0^2 - p_1^2$$

$$\text{By Entropy: } I = -p_0 \log_2(p_0) - p_1 \log_2(p_1)$$

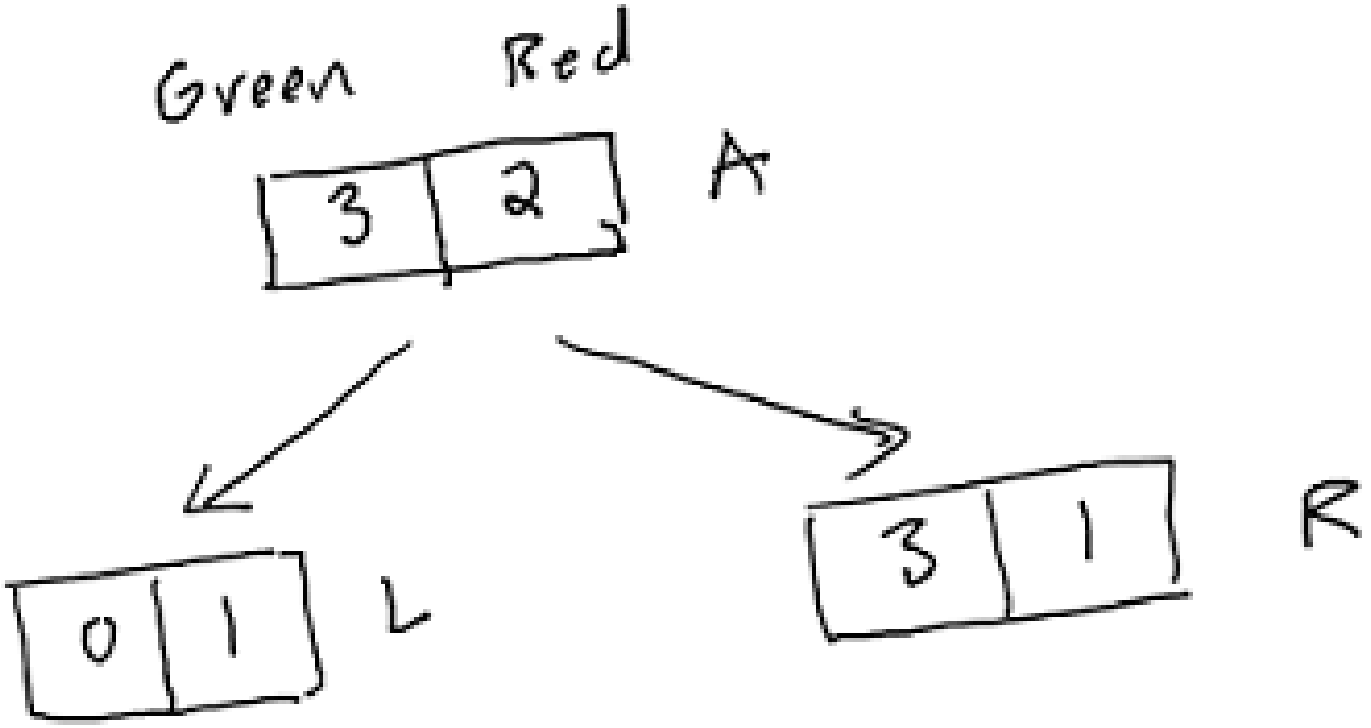
# Calculation

- Let's calculate the Children Impurity ($I_{children}$) of the three splits to decide which split is the best

# Split 1: Impurity by Classification Error

- Let **green** and **red** be class 0 and class 1, respectively.

For Split 1: $N = 5, N_{left} = 1, N_{right} = 4$



(*) By classification Error

| (neg) 0 | 1 (pos) |
|---|---|
| $p_0$ | $p_1$ |

(node A)

$$I_A = \min\{p_0, p_1\}$$

$p_0$ is porportion of 0 (neg)

$p_1$ is porportion of 1 (pos)

<u>Ex</u>

| A | | |
|---|---|---|
| | 3 | 2 |

$p_0 = \frac{3}{5}$ ; $p_1 = \frac{2}{5}$

$$\Rightarrow I_A = \min\left\{\frac{3}{5}, \frac{2}{5}\right\} = \frac{2}{5}$$

## (*) By Gini - Index

| $P_0$ | $P_1$ |
|---|---|

A

$$I_A = 1 - P_0^2 - P_1^2$$

Ex: (1)

| 3 | 2 |
|---|---|

$P_0 = 3/5$

$P_1 = 2/5$

$$\Rightarrow I_A = 1 - \left(\frac{3}{5}\right)^2 - \left(\frac{2}{5}\right)^2$$

$$I_A = \underline{\underline{.48}}$$

(2)

| 3 | 3 |
|---|---|

node B

$P_0 = P_1 = \frac{1}{2} \rightarrow I_B = 1 - \left(\frac{1}{2}\right)^2 - \left(\frac{1}{2}\right)^2$

$= 0.5$

| 7 | 0 |
|---|---|

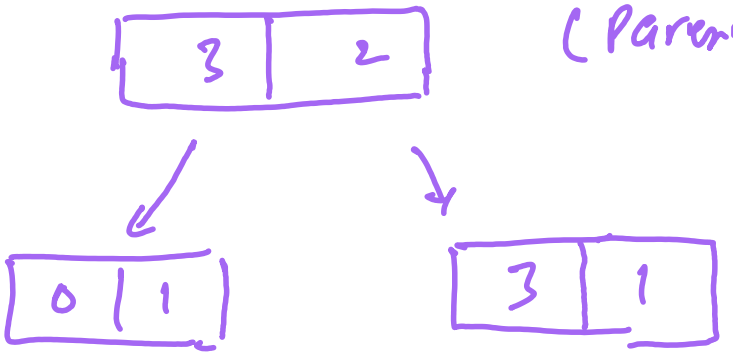
node c

$\Rightarrow \quad P_0 = \frac{7}{7} = 1$

$P_1 = 0$

$\Rightarrow \quad I_c = 1 - 1^2 - 0^2 = 0$

Split 1




(Parent)

(we are using gini-index to measure impurity)

$$I_{parent} = 1 - \left(\frac{3}{5}\right)^2 - \left(\frac{2}{5}\right)^2 = .48$$

$$I_{children} = \frac{N_{left}}{N} I_{left} + \frac{N_{right}}{N} I_{right}$$

1 4 5

$$I_{children} = \frac{1}{5}\left[1 - 0^2 - 1^2\right] + \frac{4}{5}\left[1 - \left(\frac{3}{4}\right)^2 - \left(\frac{1}{4}\right)^2\right]$$

$$= \frac{4}{5} \cdot \frac{6}{16} = .3$$

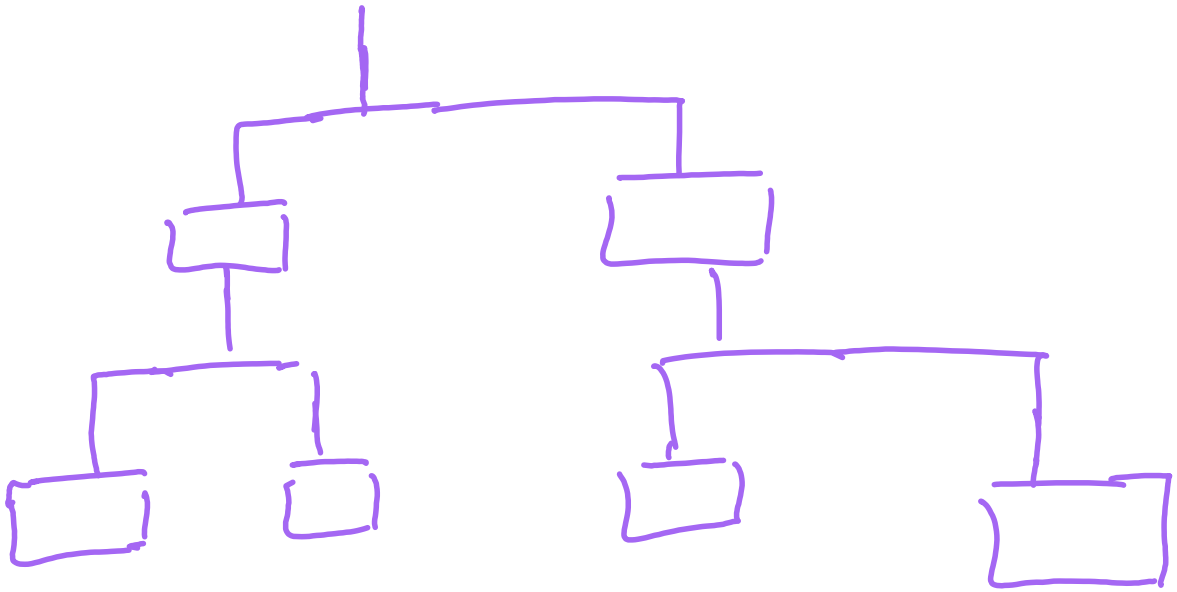
$$I_{gain} = I_{parent} - I_{children} = .48 - .3 = .18$$

split 2 $I_{gain} = .48 - .464$

$= \boxed{.016}$

split 3 $I_{gain} = .48 - .184$

$= \boxed{.296}$



# Split 1: Impurity by Classification Error

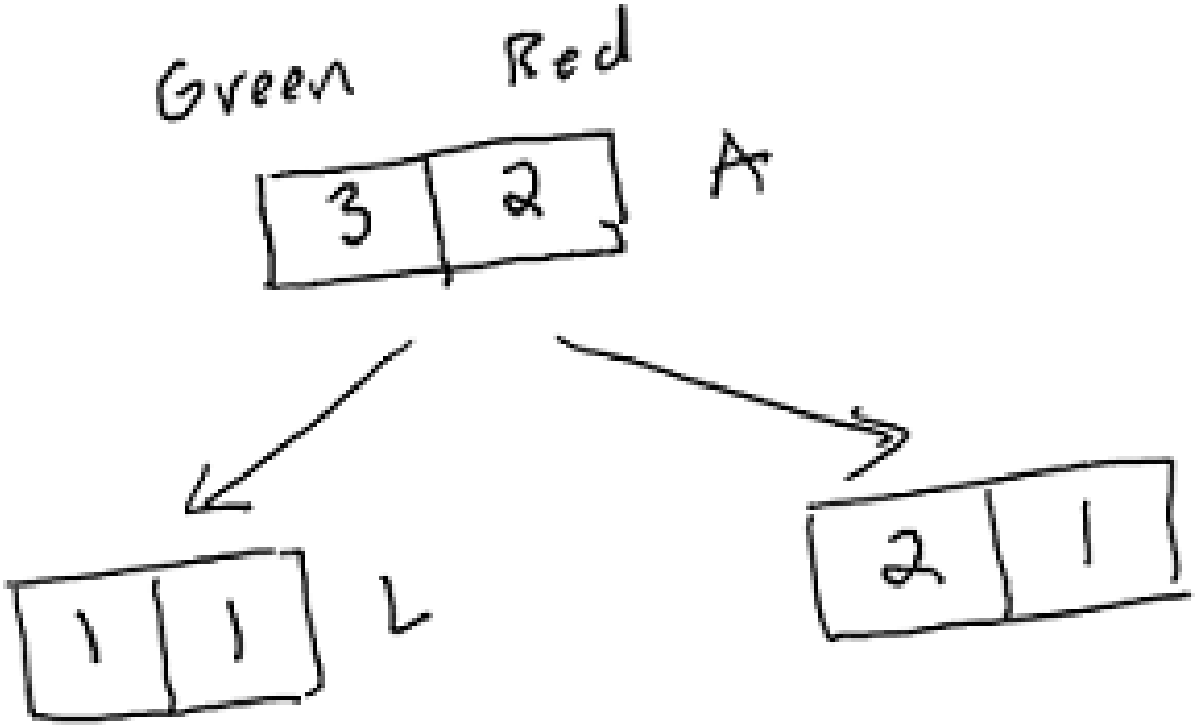
- Node *child left*, L: $p_0 = \frac{0}{1} = 0, p_1 = \frac{1}{1} = 1$. Thus, $I_L = \min(0, 1) = 0$
- Node *child right*, R: $p_0 = \frac{3}{4}, p_1 = \frac{1}{4}$. Thus, $I_R = \min(\frac{3}{4}, \frac{1}{4}) = \frac{1}{4}$
- Children Impurity of Split 1:

$$I_{children} = \frac{N_{left}}{N} I_L + \frac{N_{right}}{N} I_R$$

$$= \frac{1}{5} \cdot 0 + \frac{4}{5} \cdot \frac{1}{4} = 0.2$$

# Split 2: Impurity by Classification Error

For Split 2: $N = 5, N_{left} = 2, N_{right} = 3$



# Split 2: Impurity by Classification Error

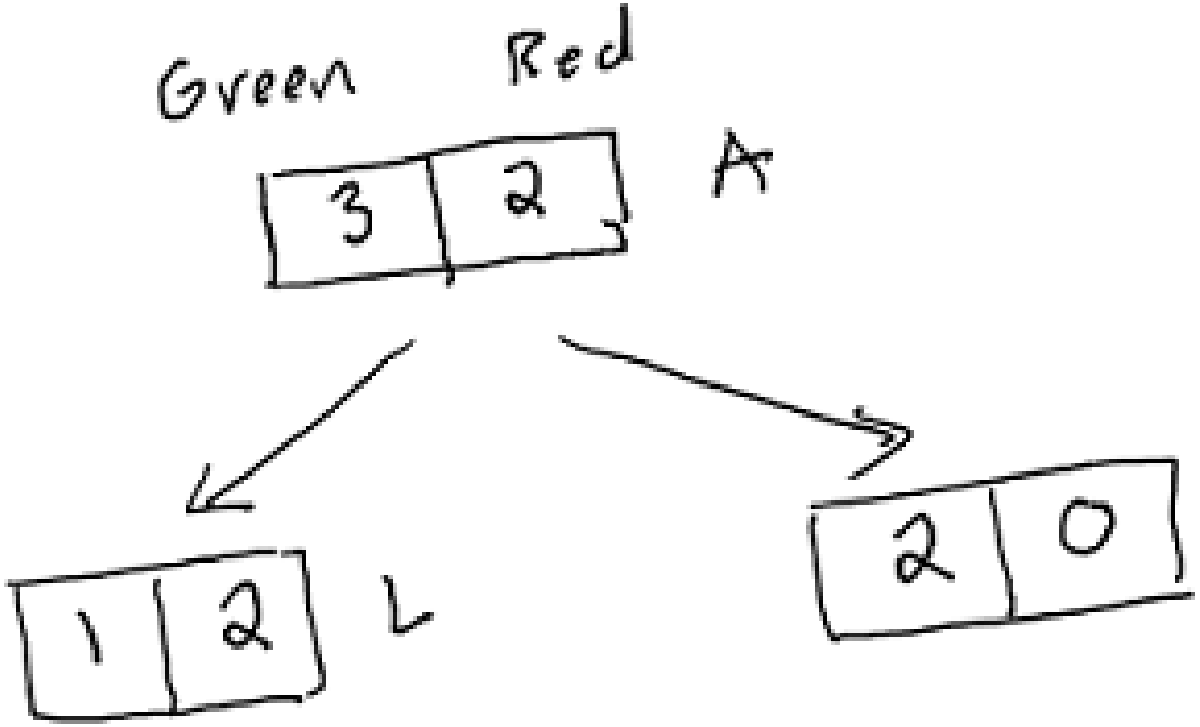
- Node *child left*, L: $p_0 = \frac{1}{2}, p_1 = \frac{1}{2}$. Thus, $I_L = \frac{1}{2}$
- Node *child right*, R: $p_0 = \frac{2}{3}, p_1 = \frac{1}{3}$. Thus, $I_R = \min(\frac{2}{3}, \frac{1}{3}) = \frac{1}{3}$
- Children Impurity of Split 2:

$$I_{children} = \frac{N_{left}}{N} I_L + \frac{N_{right}}{N} I_R$$

$$= \frac{2}{5} \cdot \frac{1}{2} + \frac{3}{5} \cdot \frac{1}{3} = 0.4$$

# Split 3: Impurity by Classification Error

For Split 3: $N = 5, N_{left} = 3, N_{right} = 2$



# Split 3: Impurity by Classification Error

- Node *child left*, L: $p_0 = \frac{1}{3}, p_1 = \frac{2}{3}$. Thus, $I_A = \min(\frac{1}{3}, \frac{2}{3}) = \frac{1}{3}$
- Node *child right*, R: $p_0 = \frac{2}{2}, p_1 = \frac{0}{2}$. Thus, $I_R = \min(1, 0) = 0$
- Children Impurity of Split 3:

$$I_{children} = \frac{N_{left}}{N} I_L + \frac{N_{right}}{N} I_R$$

$$= \frac{3}{5} \cdot \frac{1}{3} + \frac{2}{5} \cdot 0 = 0.2$$

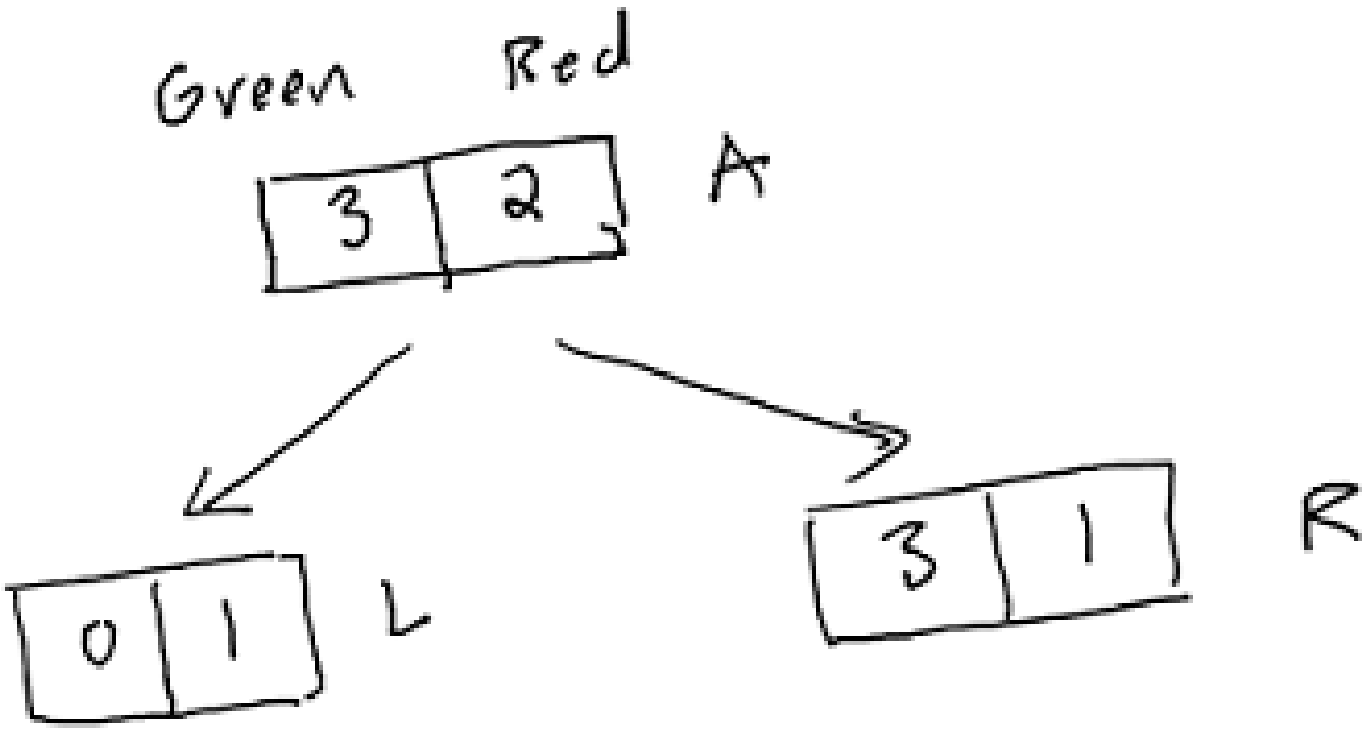
# Comparing Impurity by Classification Error

| | $I_{children}$ |
|---|---|
| Split 1 | 0.2 |
| Split 2 | 0.4 |
| Split 3 | 0.2 |

- By classification error, Split 1 and Split 3 are tie as the best because they have the same Children Impurity ($I_{children}$).

# Split 1: Impurity by Gini Index

For Split 1: $N = 5, N_{left} = 1, N_{right} = 4$



# Split 1: Impurity by Gini Index

- Node *child left*, L: $p_0 = \frac{0}{1} = 0, p_1 = \frac{1}{1} = 1$. Thus,

$$I_L = 1 - 0^2 - 1^2 = 0$$

- Node *child right*, R: $p_0 = \frac{3}{4}, p_1 = \frac{1}{4}$. Thus,

$$I_R = 1 - (\frac{3}{4})^2 - (\frac{1}{4})^2 = 0.375$$

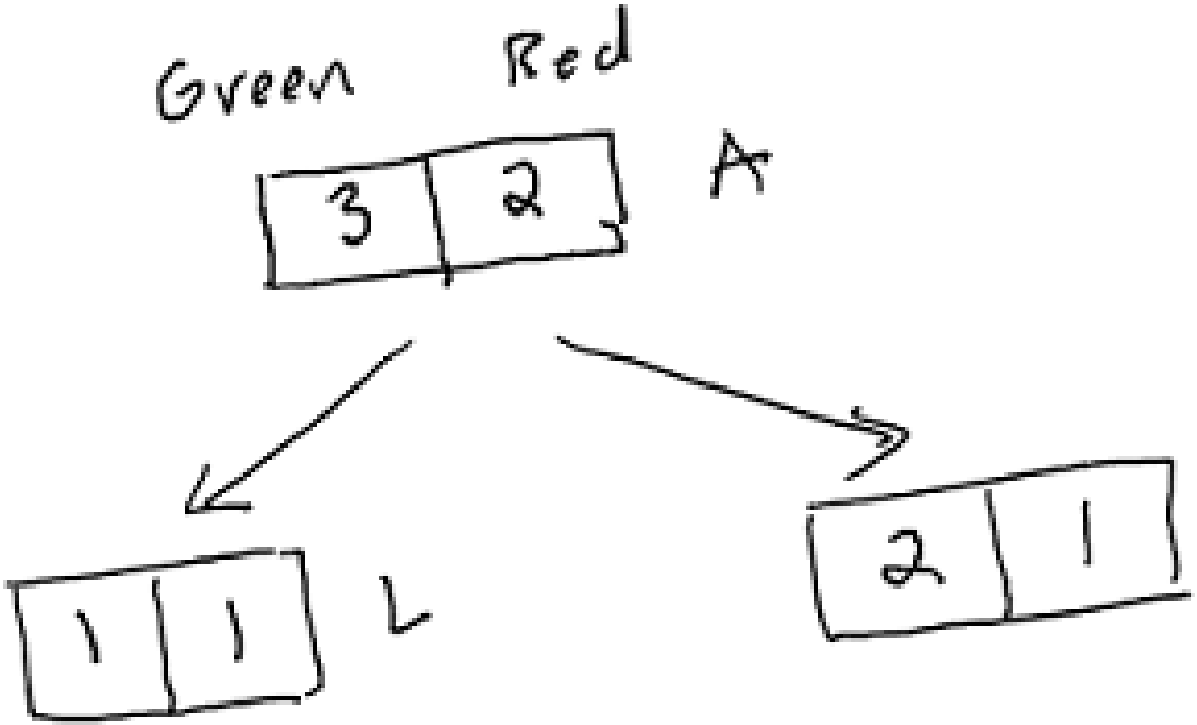
- Children Impurity of Split 1:

$$I_{children} = \frac{N_{left}}{N} I_L + \frac{N_{right}}{N} I_R$$

$$= \frac{1}{5} \cdot 0 + \frac{4}{5} \cdot 0.375 = 0.3$$

# Split 2: Impurity by Gini Index

For Split 2: $N = 5, N_{left} = 2, N_{right} = 3$



# Split 2: Impurity by Gini Index

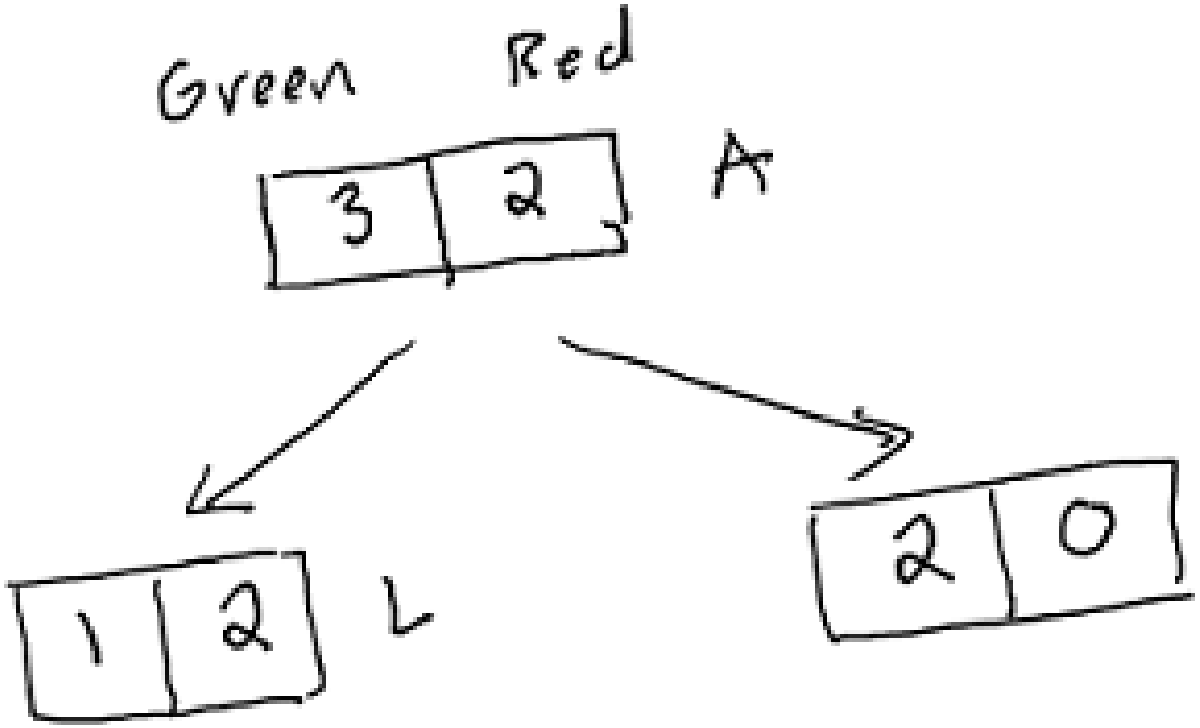
- Node *child left*, L: $p_0 = \frac{1}{2}, p_1 = \frac{1}{2}$. Thus, $I_L = 1 - (\frac{1}{2})^2 - (\frac{1}{2})^2 = 0.5$
- Node *child right*, R: $p_0 = \frac{2}{3}, p_1 = \frac{1}{3}$. Thus, $I_R = 1 - (\frac{2}{3})^2 - (\frac{1}{3})^2 = 0.44$
- Children Impurity of Split 2:

$$I_{children} = \frac{N_{left}}{N} I_L + \frac{N_{right}}{N} I_R$$

$$= \frac{2}{5} \cdot \frac{1}{2} + \frac{3}{5} \cdot 0.44 = \underline{0.464}$$

# Split 3: Impurity by Gini Index

For Split 3: $N = 5, N_{left} = 3, N_{right} = 2$



# Split 3: Impurity by Gini Index

- Node *child left*, L: $p_0 = \frac{1}{3}, p_1 = \frac{2}{3}$. Thus, $I_A = 1 - (\frac{1}{3})^2 - (\frac{2}{3})^2 = 0.44$
- Node *child right*, R: $p_0 = \frac{2}{2}, p_1 = \frac{0}{2}$. Thus, $I_R = 1 - 0^2 - 1^2 = 0$
- Children Impurity of Split 3:

$$I_{children} = \frac{N_{left}}{N} I_L + \frac{N_{right}}{N} I_R$$

$$= \frac{3}{5} \cdot 0.44 + \frac{2}{5} \cdot 0 = \boxed{0.184}$$

# Comparing Impurity by Gini Index

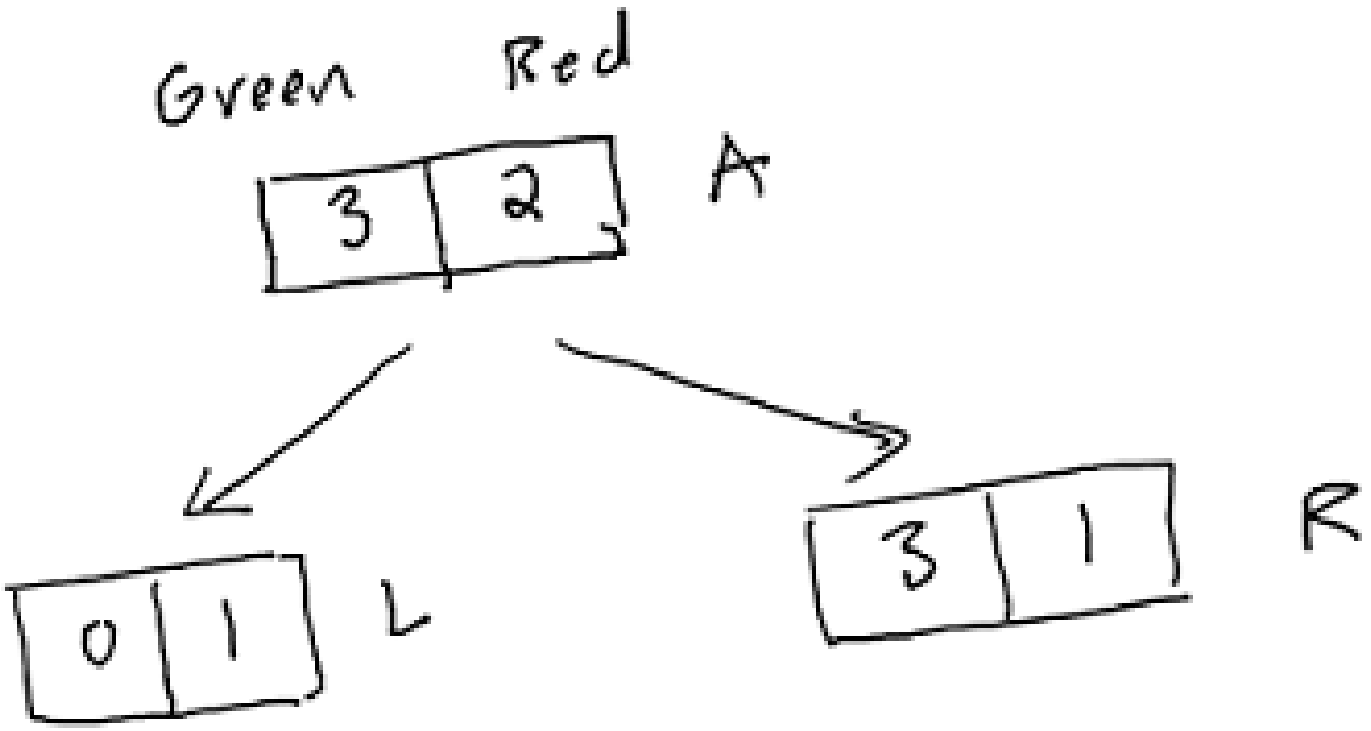
| | $I_{children}$ |
|---|---|
| Split 1 | 0.3 |
| Split 2 | 0.464 |
| Split 3 | 0.184 |

best split
( lowest impurity )

- By Gini Index, Split 3 is the best because it has the smallest Children Impurity ($I_{children}$).

# Split 1: Impurity by Entropy

For Split 1: $N = 5, N_{left} = 1, N_{right} = 4$



# Split 1: Impurity by Entropy

- Node *child left*, L: $p_0 = \frac{0}{1} = 0, p_1 = \frac{1}{1} = 1$. Thus, $I_L = 0$
- Node *child right*, R: $p_0 = \frac{3}{4}, p_1 = \frac{1}{4}$. Thus,

$$I_R = -log_2(\frac{3}{4}) - log_2(\frac{1}{4}) = 0.811$$

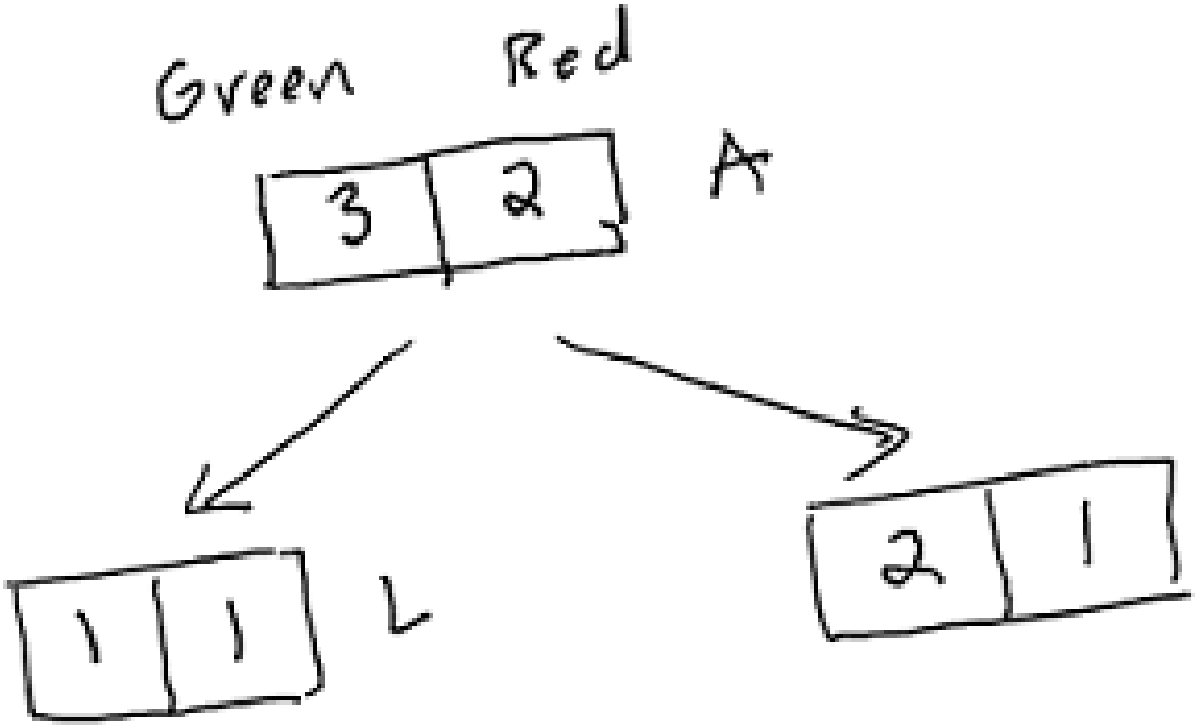
- Children Impurity of Split 1:

$$I_{children} = \frac{N_{left}}{N} I_L + \frac{N_{right}}{N} I_R$$

$$= \frac{1}{5} \cdot 0 + \frac{4}{5} \cdot 0.811 = 0.649$$

# Split 2: Impurity by Entropy

For Split 2: $N = 5, N_{left} = 2, N_{right} = 3$



# Split 2: Impurity by Entropy

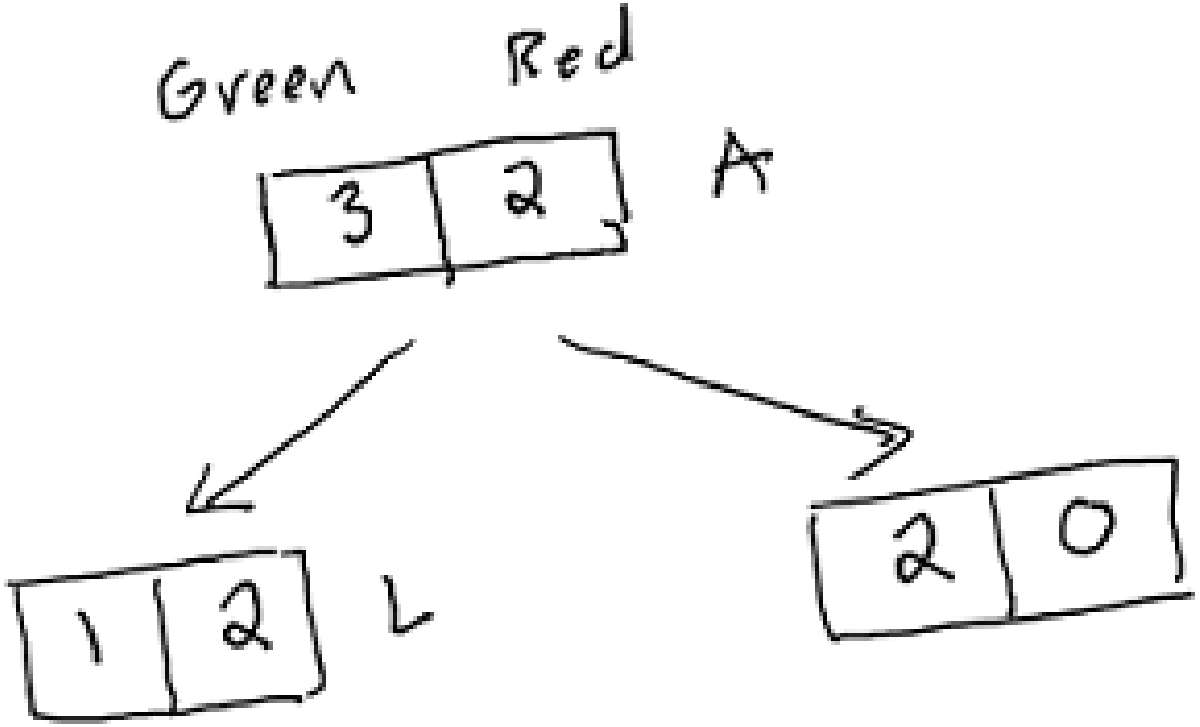
- Node *child left*, L: $p_0 = \frac{1}{2}, p_1 = \frac{1}{2}$. Thus, $I_L = -log_1(\frac{1}{2}) - log_2(\frac{1}{2}) = 1$
- Node *child right*, R: $p_0 = \frac{2}{3}, p_1 = \frac{1}{3}$. Thus, $I_R = -log_2(\frac{2}{3}) - log_2(\frac{1}{3}) = 0.918$
- Children Impurity of Split 2:

$$I_{children} = \frac{N_{left}}{N} I_L + \frac{N_{right}}{N} I_R$$

$$= \frac{2}{5} \cdot 1 + \frac{3}{5} \cdot 0.918 = 0.951$$

# Split 3: Impurity by Entropy

For Split 3: $N = 5, N_{left} = 3, N_{right} = 2$



# Split 3: Impurity by Entropy

- Node *child left*, L: $p_0 = \frac{1}{3}, p_1 = \frac{2}{3}$. Thus, $I_A = -log_2(\frac{1}{3}) - log_2(\frac{2}{3}) = 0.918$
- Node *child right*, R: $p_0 = \frac{2}{2}, p_1 = \frac{0}{2}$. Thus, $I_R = 0$
- Children Impurity of Split 3:

$$I_{children} = \frac{N_{left}}{N} I_L + \frac{N_{right}}{N} I_R$$

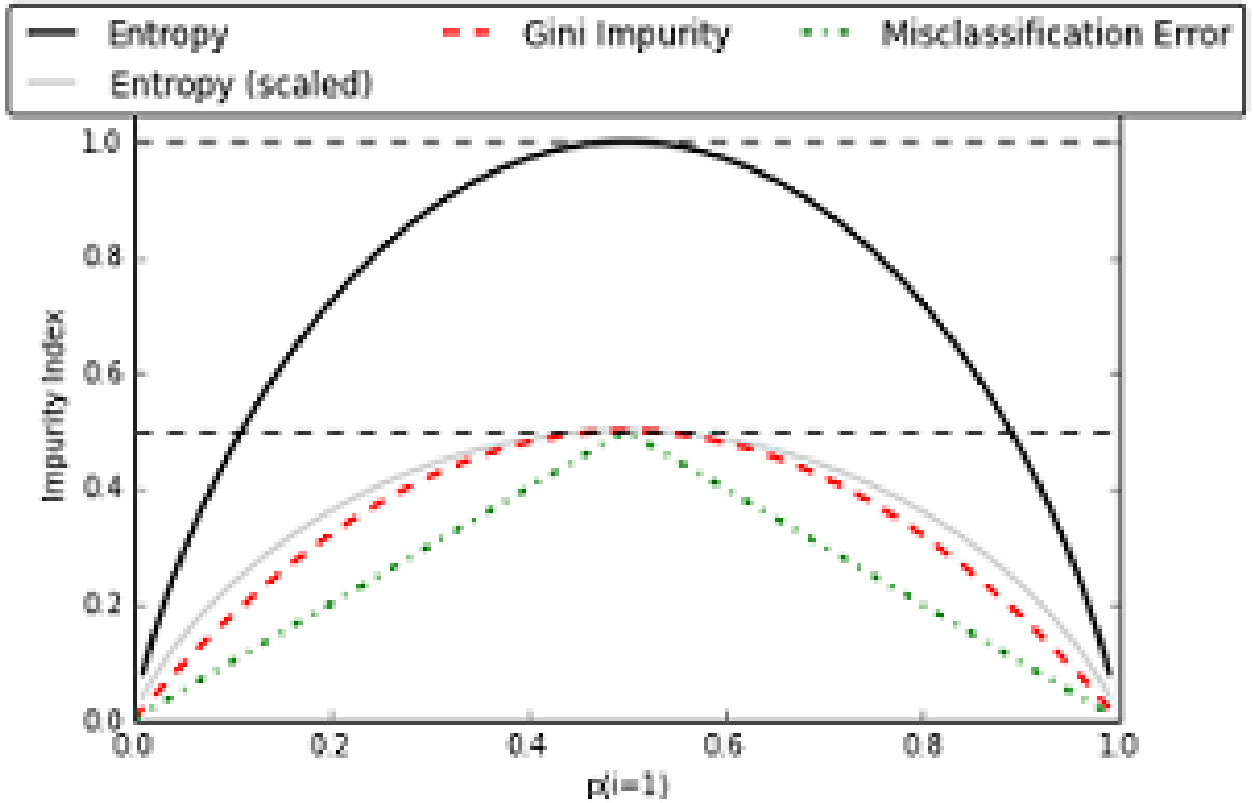
$$= \frac{3}{5} \cdot 0.918 + \frac{2}{5} \cdot 0 = 0.551$$

# Comparing Impurity by Entropy

| | $I_{children}$ |
|---|---|
| Split 1 | 0.649 |
| Split 2 | 0.951 |
| Split 3 | 0.551 |

- By Gini Index, Split 3 is the best because it has the smallest Children Impurity ($I_{children}$).

# Comparing Impurity Measures

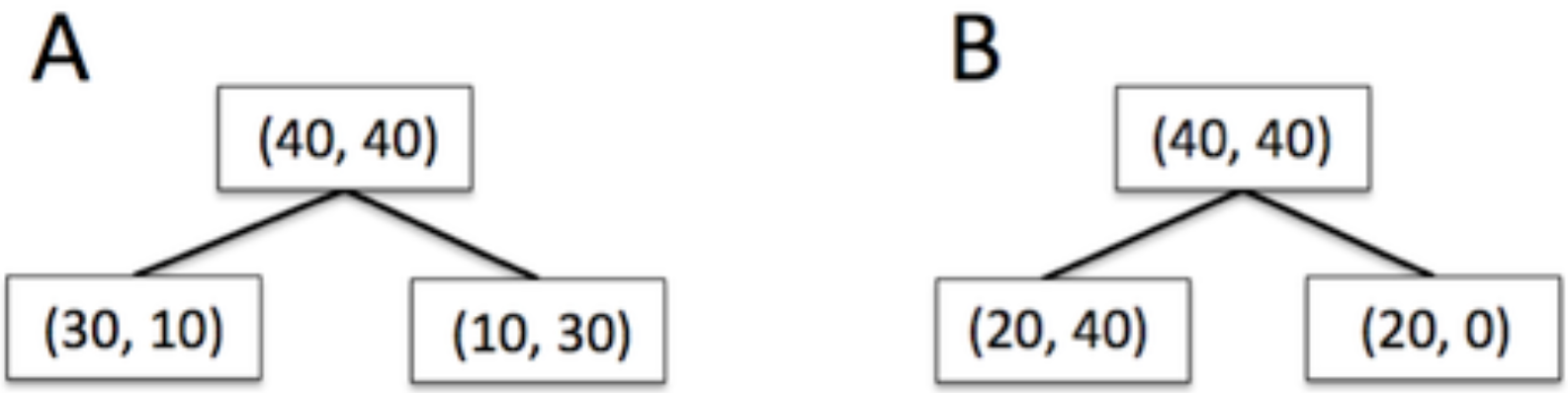



- Relation between impurity and the class probabilities. All impurity measures are maximized at $p_1 = 1/2$ and minimized at $p_1 = 0$ and $p_1 = 1$.

# Another Example

( submit to canvas for extra credit)

calculating using entropy or gini_index or classification errors)



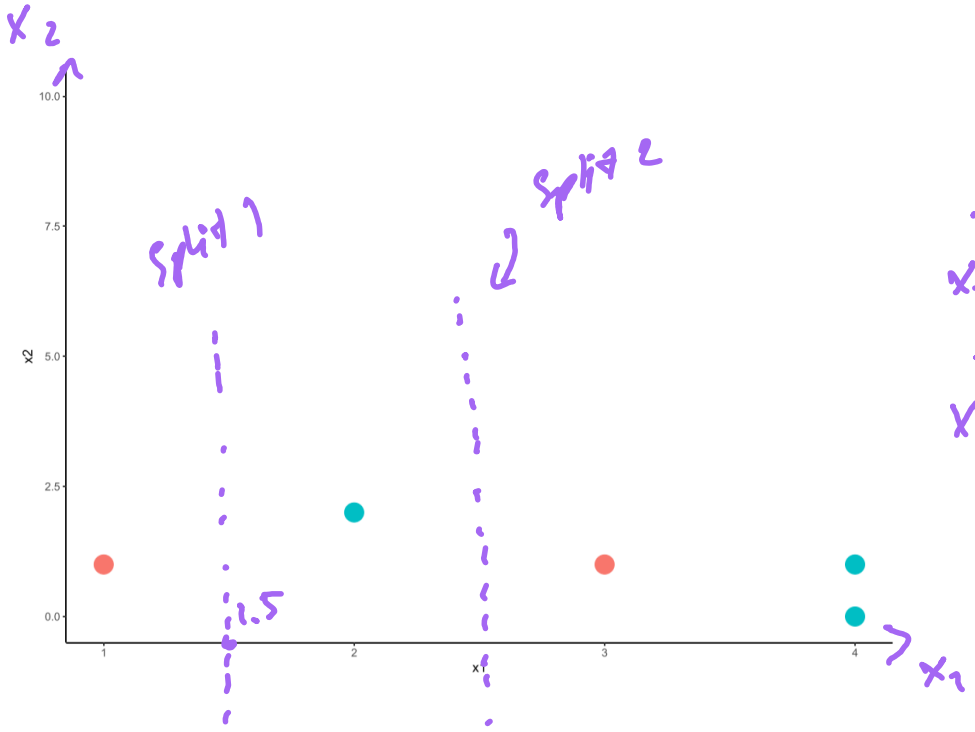

- Which split is better?

visually, which one is better ?

A better : 8

B better : 1

same : 0




split 1

| | Red | Green |
|---|---|---|
| $x_1 \geq 1.5$ | 1 | 3 |
| $x_1 < 1.5$ | 1 | 0 |

split 2

| | Red | Green |
|---|---|---|
| $x_1 \geq 2.5$ | 1 | 2 |
| $x_1 < 2.5$ | 1 | 1 |

$V_3$

$V_4$

⊕ **Variable 1** (binary variable) → Red, Green

⊕ **variable 2** (binary) → yes ($x_1 \geq 1.5$), no (not $\geq 1.5$)

Gender Fav music Majors

# Decide the best split using Chi-Square test of Independence

- Besides Children Impurity, one can use the Chi-square, $\chi^2$, test of independence to decide the best split.

# Review of Chi-Square test of Independence

- Let $X$ and $Y$ be two categorical variables.
- We want to test if $X$ and $Y$ are independent/associated
  - $H_0$: $X$ and $Y$ are independent
  - $H_\alpha$ : $X$ and $Y$ are dependent
- Test statistic:

$$\sum \frac{(e_i - o_i)^2}{e_i} \sim \chi^2 \text{ distribution with degree of freedom}(n-1)(m-1)$$

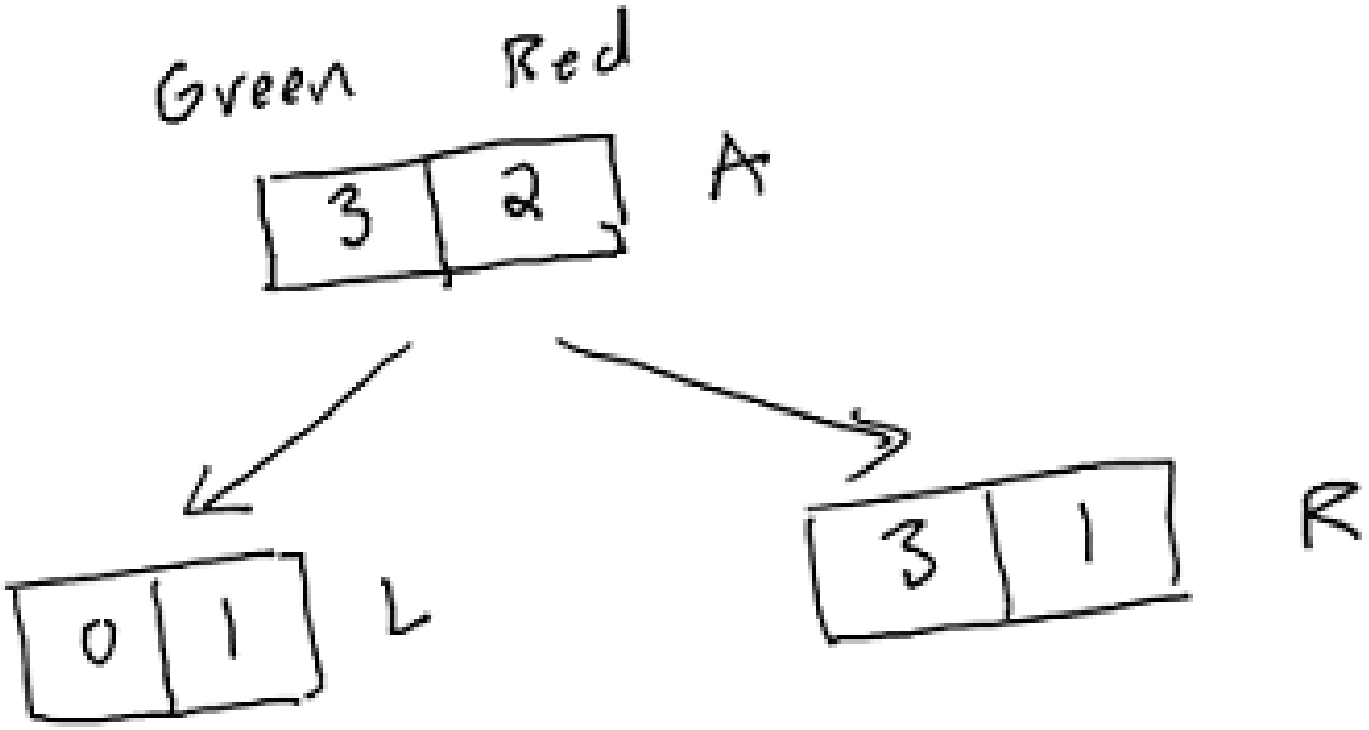
# Review of Chi-Square test of Independence

- In our context, the greater the $\chi^2$ value, the smaller the $p - value$
- The smaller the $p - value$, the more dependent the two variables are. Thus the better the split is.
- Therefore, we look for the split with the **greatest $\chi^2$ value.**

# Applying to Our Example

- We will calculate the $\chi^2$ values of the three splits.
- The best split is the split with the greatest $\chi^2$ value.

# Split 1




| | Greens | Reds | Total |
|---|---|---|---|
| Left Branch | 0 | 1 | 1 |
| Right Branch | 3 | 1 | 4 |
| Total | 3 | 2 | |

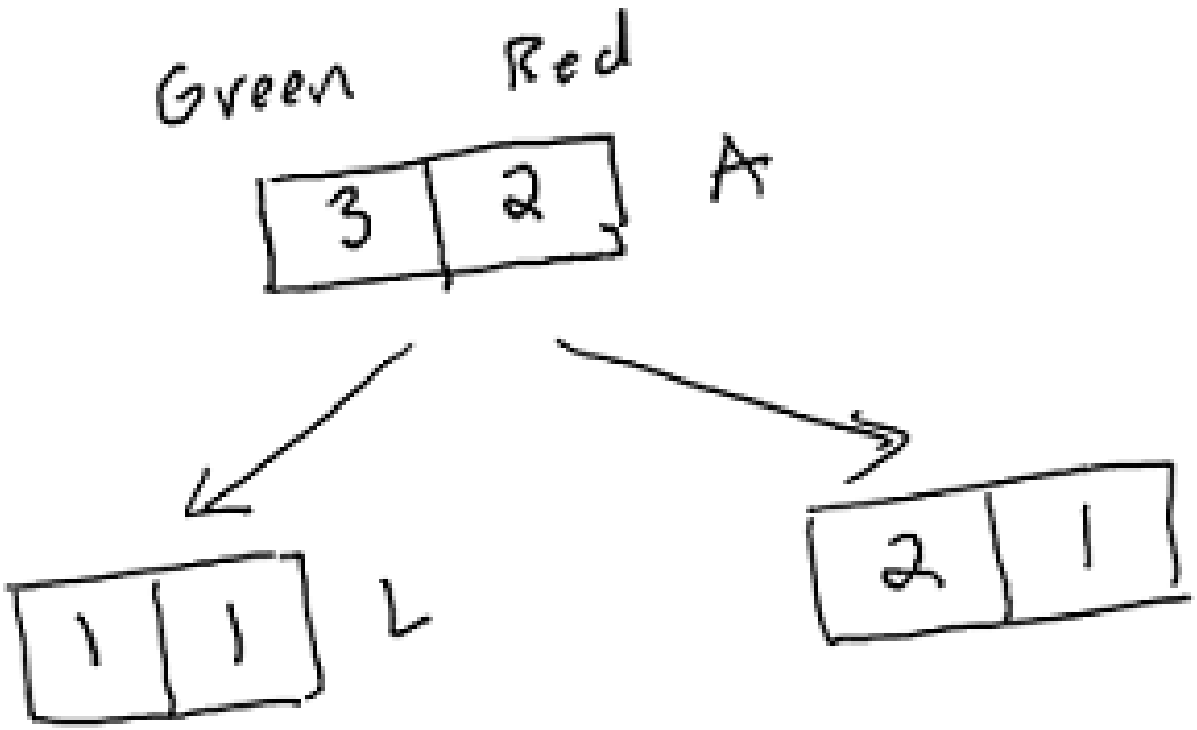
# Split 1

$$\chi^2 = \frac{(e_1 - o_1)^2}{e_1} + \frac{(e_2 - o_2)^2}{e_2} + \frac{(e_3 - o_3)^2}{e_3} + \frac{(e_4 - o_4)^2}{e_4}$$

- $i = 1$ (Cell 1): $e_1 = \frac{1 \cdot 3}{5}$, $o_1 = 0$
- $i = 2$ (Cell 2): $e_2 = \frac{1 \cdot 2}{5}$, $o_2 = 1$
- $i = 3$ (Cell 3): $e_3 = \frac{3 \cdot 4}{5}$, $o_3 = 3$
- $i = 4$ (Cell 4): $e_4 = \frac{2 \cdot 4}{5}$, $o_4 = 1$
- Plug in, we have:

$$\chi^2 = 1.875$$

# Split 2




| | Greens | Reds | Total |
|---|---|---|---|
| Left Branch | 1 (Cell 1) | 1 (Cell 2) | 2 |
| Right Branch | 2 (Cell 3) | 1 (Cell 4) | 3 |
| Total | 3 | 2 | |

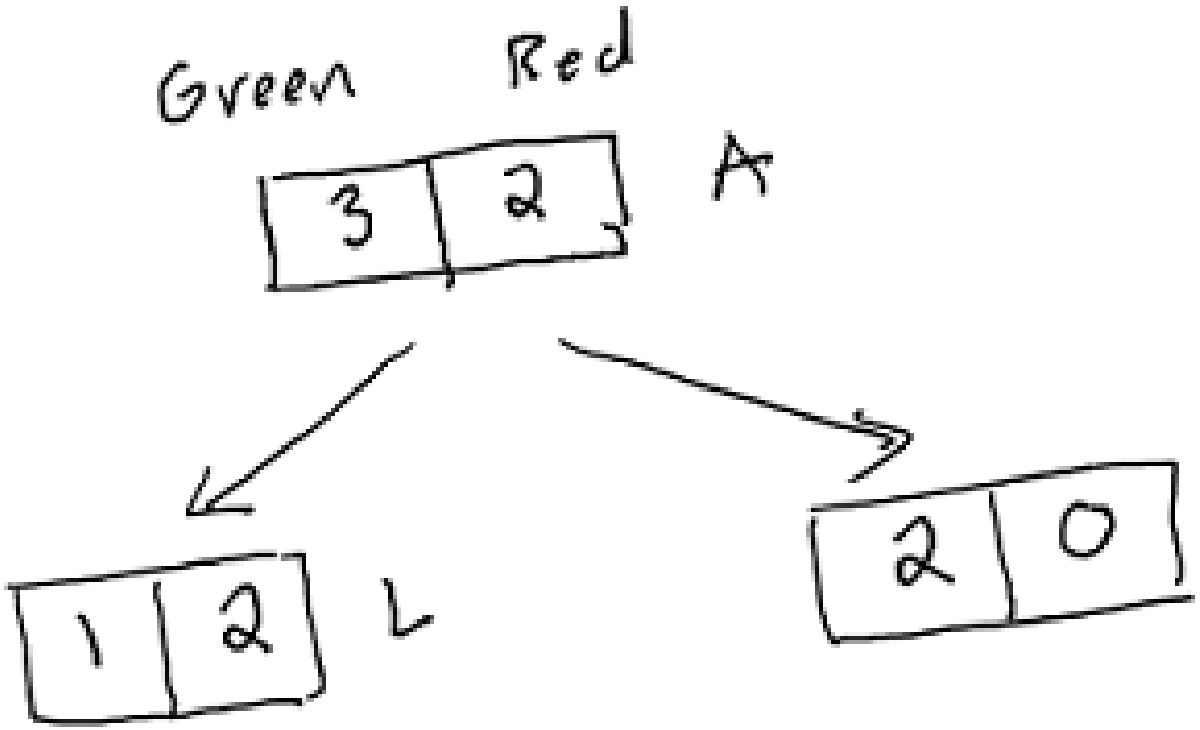
# Split 2

$$\chi^2 = \frac{(e_1 - o_1)^2}{e_1} + \frac{(e_2 - o_2)^2}{e_2} + \frac{(e_3 - o_3)^2}{e_3} + \frac{(e_4 - o_4)^2}{e_4}$$

- $i = 1$ (Cell 1): $e_1 = \frac{2 \cdot 3}{5}$, $o_1 = 1$
- $i = 2$ (Cell 2): $e_2 = \frac{2 \cdot 2}{5}$, $o_2 = 1$
- $i = 3$ (Cell 3): $e_3 = \frac{3 \cdot 3}{5}$, $o_3 = 2$
- $i = 4$ (Cell 4): $e_4 = \frac{3 \cdot 2}{5}$, $o_4 = 1$
- Plug in, we have:

$$\chi^2 = 0.139$$

# Split 3




| | Greens | Reds | Total |
|---|---|---|---|
| Left Branch | 1 (Cell 1) | 2 (Cell 2) | 3 |
| Right Branch | 2 (Cell 3) | 0 (Cell 4) | 2 |
| Total | 3 | 2 | |

# Split 3

$$\chi^2 = \frac{(e_1 - o_1)^2}{e_1} + \frac{(e_2 - o_2)^2}{e_2} + \frac{(e_3 - o_3)^2}{e_3} + \frac{(e_4 - o_4)^2}{e_4}$$

- (Cell 1): $e_1 = \frac{2 \cdot 3}{5}$, $o_1 = 1$
- (Cell 2): $e_2 = \frac{2 \cdot 2}{5}$, $o_2 = 2$
- (Cell 3): $e_3 = \frac{3 \cdot 3}{5}$, $o_3 = 2$
- (Cell 4): $e_4 = \frac{3 \cdot 2}{5}$, $o_4 = 0$
- Plug in, we have:

$$\chi^2 = 2.222$$

# Comparing the three splits

| | $\chi^2$ |
|---|---|
| Split 1 | 1.875 |
| Split 2 | 0.139 |
| Split 3 | 2.222 |

- Split 3 is the best because it has the greatest $\chi^2$!

# Logworth

- The quality of the split can be measured by **Logworth**
- Formula:

$$logworth = -log(p_{value})$$

- The greater the logworth, the better the split

# Logworth

| | $\chi^2$ | p-value | logworth |
|---|---|---|---|
| Split 1 | 1.875 | 0.114 | 0.943 |
| Split 2 | 0.139 | 0.998 | 0.0008 |
| Split 3 | 2.222 | 0.088 | 1.055 |

- Greatest $\chi^2$ = Lowest $p - value$ = Greatest logworth = Best Split
- Split 3 is the best split!

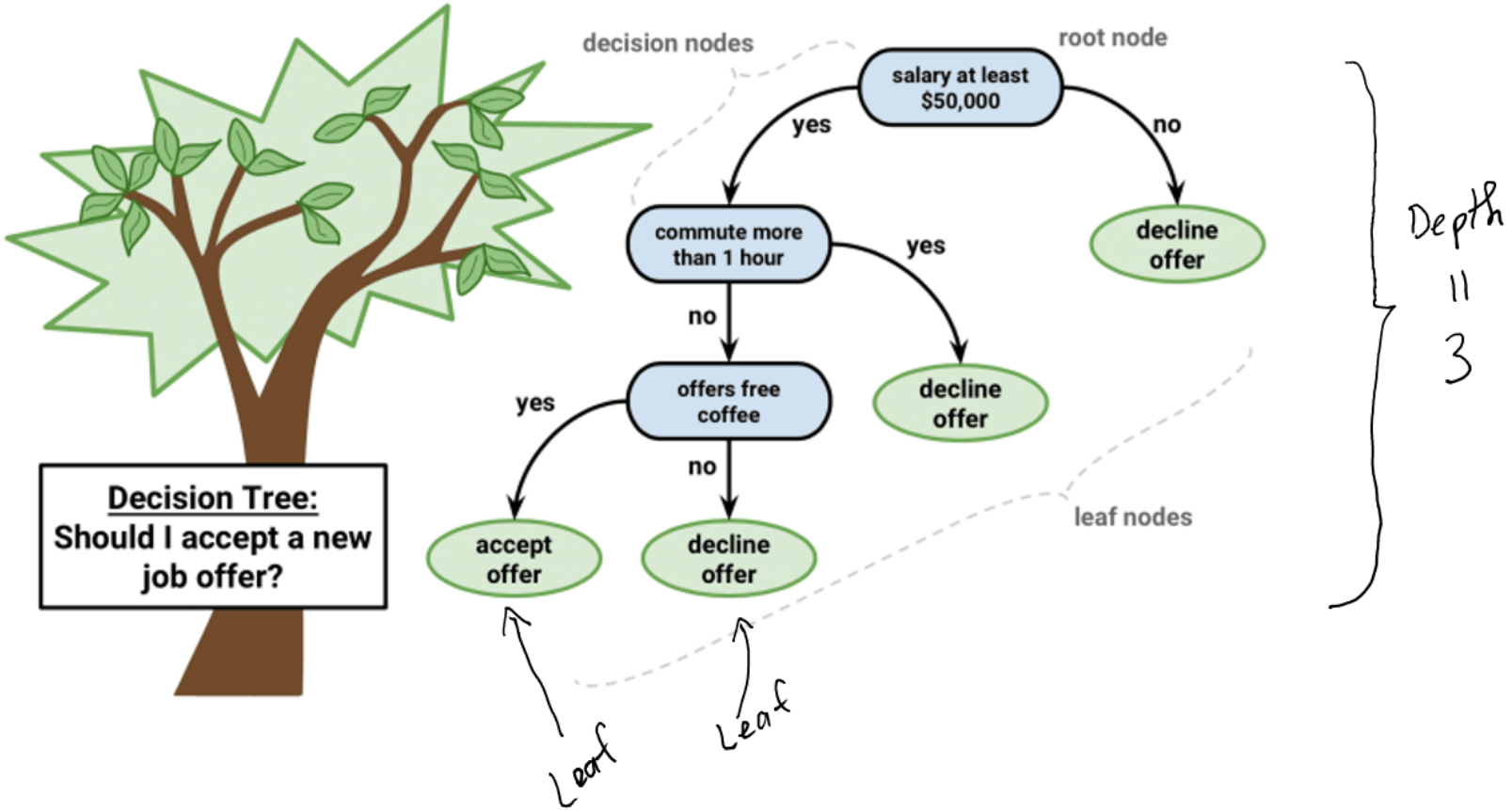
# What happens after the first split?

- After the first split, the data are divided into to subsets.
- The splitting process is repeated for each subset.
- The process ends when a stopping criteria is satisfied

# Stopping Criteria

- Minimum Leaf Size: The minimum of observations in the leaves
- Maximum Number of Leaves
- Maximum Depth
- Others

# Stopping Criteria



# Decision Tree Algorithm - How to grow a tree

- Step 1: Calculate the Children Impurity or $p - value$ of all possible splits at all variables
- Step 2: Select the split that give the minimum Children Impurity or lowest $p - value$ to split the data into two subdata $D_1$ and $D_2$
- Repeat *Step 1* and *Step 2* to both $D_1$ and $D_2$.
- Until a stopping criteria is satisfied

# Complexity of Decision Tree

- A complexity of a tree can be measured by the number of leaves the tree has
- The more leaves a tree has, the more complex the tree is.
- A complex tree may be **overfitted**, i.e. having low training error but high testing error.

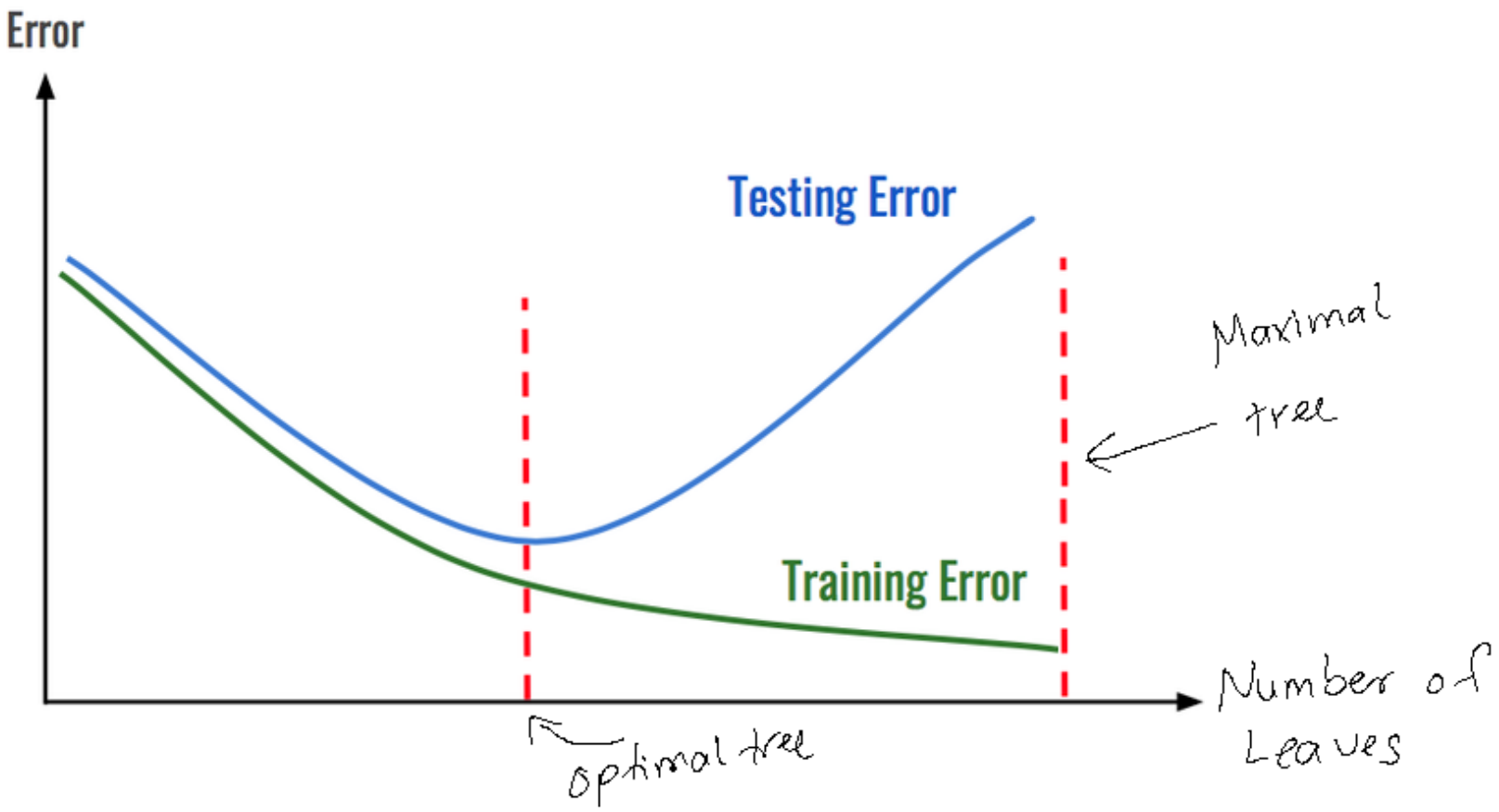
# Prunning a tree

- For any given data, one can construct a tree that achives 0 misclassification on training data
- After growing the tree one needs to prune it to avoid overfittted

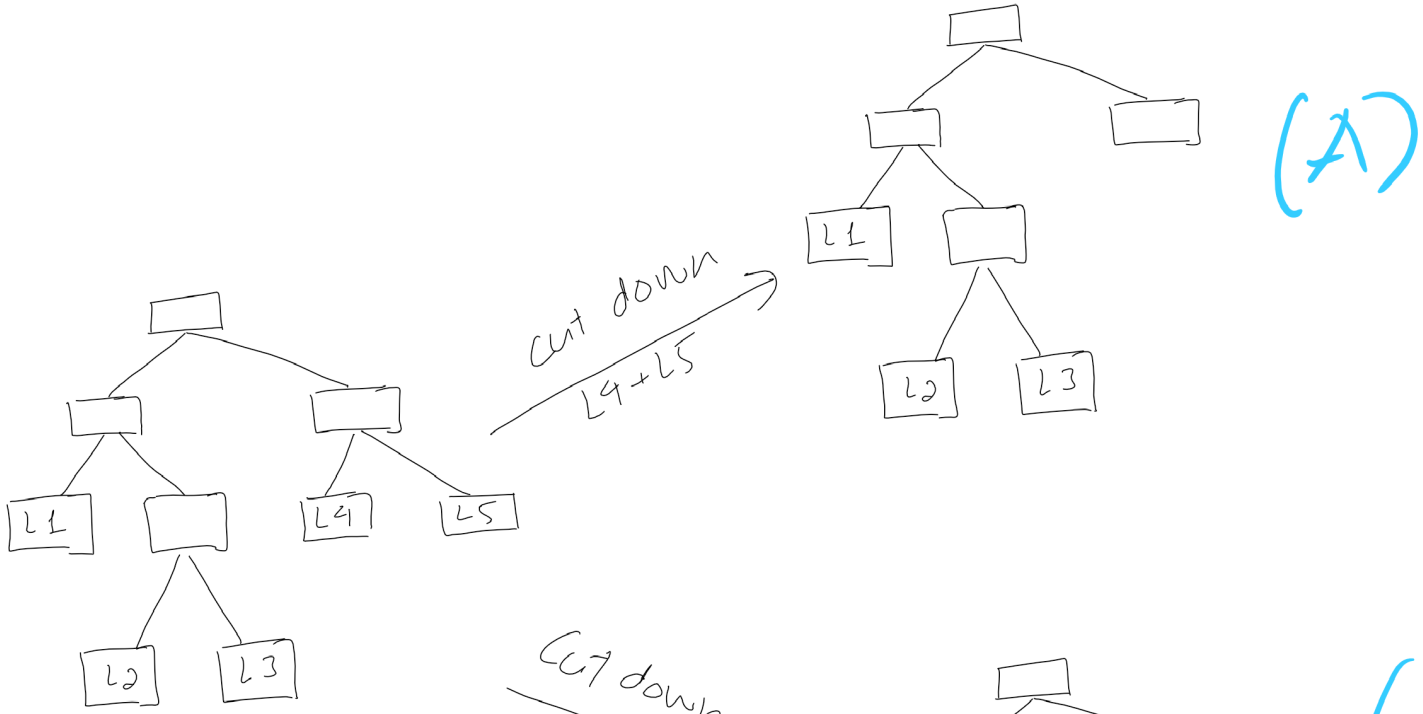
# Prunning a tree

- The tree with maximum number of leaves is called the **maximal tree** (still satisfied the stopping rule)
- From the **maximal tree**, leaves are cut down, one by one, to obatined all possible subtrees
- The subtree with lowest error on validation data, is the **optimal tree**

# Maximal vs Optimal Tree



# Example of Tree Prunning



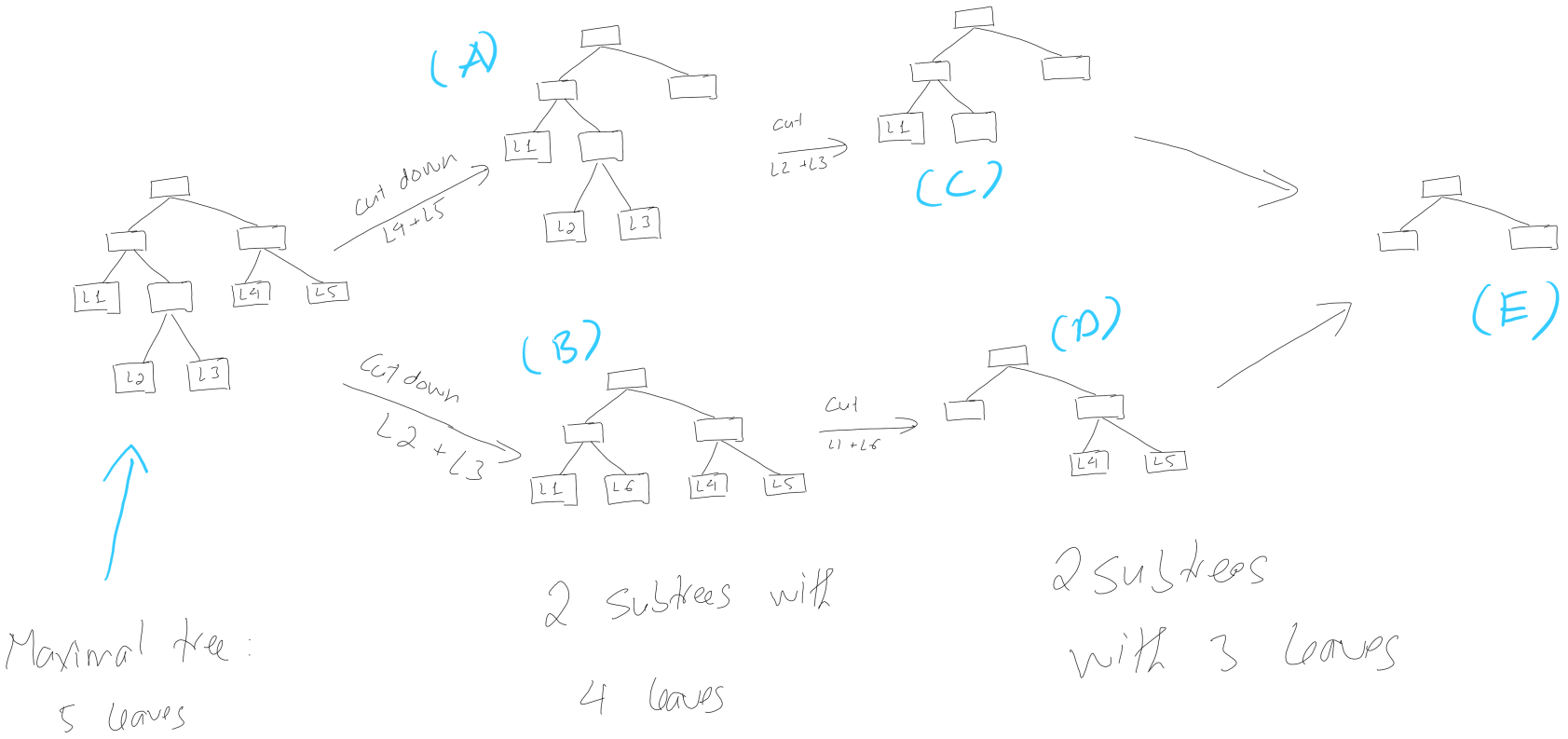

Maximal tree:

5 leaves

2 subtrees with

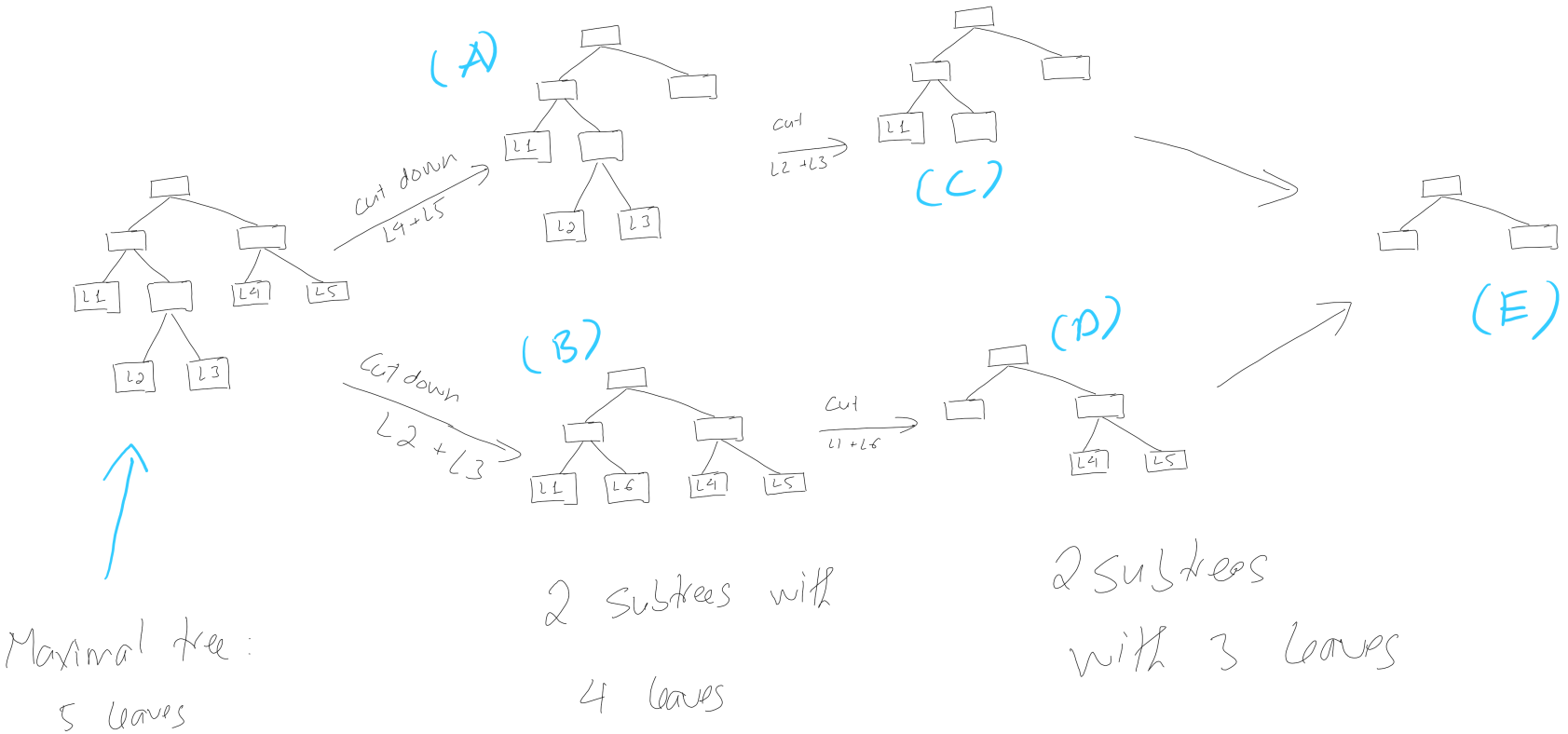
4 leaves

# Example of Tree Prunning




- All the subtrees A, B, C, D, and E will be validated with the validation data to find the **optimal tree**
- The **optimal tree** could be the **maximal tree**!

# Question




- What if both B and C give the lowest error on the validation data? Which tree should be selected as the final model?